



# Global monetary policy shocks and the adaptation of supply chains

Chang Li <sup>a,1,3</sup>, Wei Li <sup>b,2,3,\*</sup>, Yuhui Shao <sup>c,3</sup>

<sup>a</sup> School of Economics and Management, East China Normal University, China

<sup>b</sup> School of Insurance and Economics, University of International Business and Economics, China

<sup>c</sup> China COSCO SHIPPING Corporation Limited, China

## ARTICLE INFO

### JEL classification:

E50  
F23  
L14

### Keywords:

Monetary policy  
Supply chains  
Disruption  
Supplier switch

## ABSTRACT

This paper examines how global monetary policy shocks (MPS) transmitted through trade networks affect supply chain adaptation of Chinese firms. When exposed to higher supplier-side MPS, firms experience significantly higher separation rates and lower foreign entry rates. Our findings remain robust across a range of alternative MPS measures. **In response, firms shift their sourcing from high-interest-rate countries to low-interest-rate countries, but they do not re-shore.** We show that trade credit and price adjustment by suppliers are important for explaining the restructuring. The **disruption** is more likely driven by customer decisions. Moreover, heterogeneities show that firms with stricter credit constraints are more sensitive to MPS, **while longer-duration supply chain relationships show greater resilience.** The supply chain adaptation induced by MPS also leads to adverse real economic outcomes for imports, exports, and profitability.

## 1. Introduction

Firms always face pressure to organize their production through complex but resilient supply chains. Despite its importance, supply chain disruptions have become the new normal (Grossman et al., 2023). In the past decades, many multinational firms have well-established global supply chains to exploit the comparative advantage of partner countries (Klaus et al., 2020; Minetti et al., 2019). Global supply chains are inevitably subject to economic, weather, and political shocks in foreign markets, such as natural disasters (Vasco et al., 2021; Pamina & Poncet, 2022), unanticipated tariff (Gene et al., 2024), turnover of political leaders (Chuoyan et al., 2022), and pandemic lockdowns (Khanna et al., 2022), among others, which pressure firms to consider multiple tradeoffs and restructure supply chains.

Despite that the conditions of the financial sector have a large and persistent effect on firm operations, much less is known about how firms adapt their global supply chains in response to monetary policy shocks in partner countries. Monetary policy largely determines aggregate capital availability and the costs of financing (Ben & Blinder, 1992), which have sizable direct effects on firm investment decisions and efficiency. A deterioration of funding conditions reduces the total supply of goods, induces price adjustments, or changes the patterns of trade credit, and the shocks can ripple through supply chains (Alfaro et al., 2021; Chen, 2023; Anna, 2020).

\* Corresponding author.

E-mail addresses: [cli@fem.ecnu.edu.cn](mailto:cli@fem.ecnu.edu.cn) (C. Li), [weili.uibe@gmail.com](mailto:weili.uibe@gmail.com) (W. Li), [yhshao2020@nsd.pku.edu.cn](mailto:yhshao2020@nsd.pku.edu.cn) (Y. Shao).

<sup>1</sup> Chang Li acknowledges the financial support from the National Natural Science Foundation of China No. 72341012, No. 72433002, and No. 72303066.

<sup>2</sup> Wei Li acknowledges the financial support from the National Natural Science Foundation of China No. 72303067, the Shanghai Pujiang Program No. 23PJJC032, and the Shanghai Philosophy and Social Sciences Young Scholar Program No. 2023EJB009.

<sup>3</sup> All authors contributed equally to this research.

<https://doi.org/10.1016/j.chieco.2025.102535>

Received 5 August 2024; Received in revised form 27 July 2025; Accepted 11 August 2025

Available online 23 August 2025

1043-951X/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

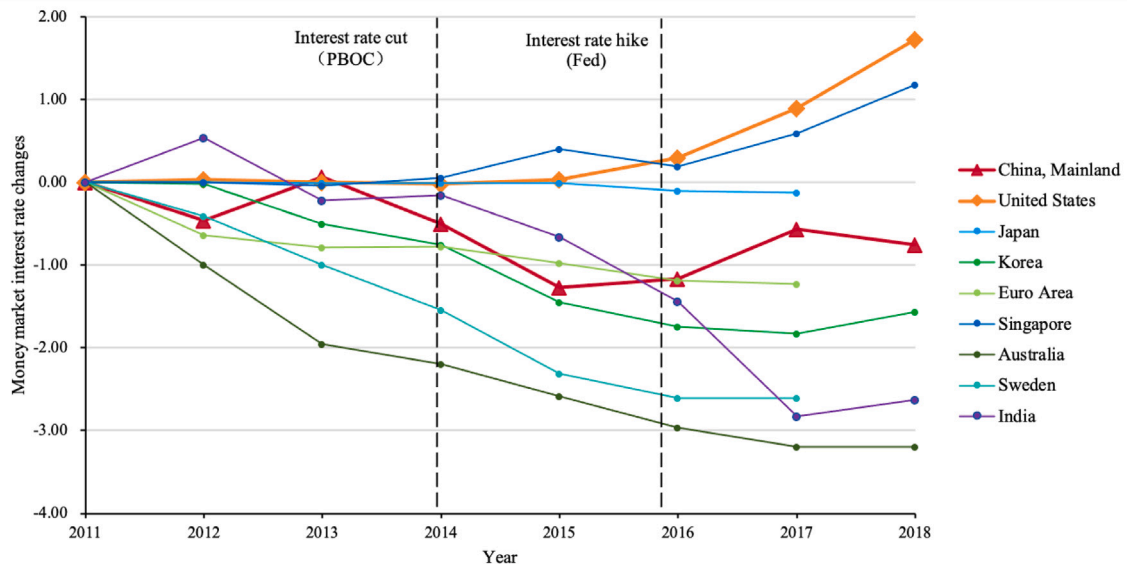


Fig. 1. Changes in the Money Market Interest Rates of Major Supplier Countries.

This figure presents the changes in the money market interest rate of the major supplying countries for Chinese listed firms. To facilitate comparison, we use the money market interest rate of each country in the year 2011 as the base and present the changes in money market interest rates in the following years.

In a globalized economy, suppliers are distributed worldwide. Countries typically have their own priorities for economic development and implement inconsistent monetary policies. Fig. 1 presents the money market interest rates of the major economic entities, exhibiting significant divergences in interest rate dynamics around the world. As a result, multinational firms are exposed to unanticipated monetary policy shocks, in terms of either magnitude or direction, from their suppliers. For example, the Federal Reserve (Fed) has raised the fed funds rate several times since 2022. While some economies, like South Korea and the EU, followed the Fed by implementing interest rate hikes, others, like China and Japan, maintained relatively stable interest rates. **These opposing policies have likely contributed to the collapse of Korea's shipbuilding industry and the rapid growth of China's shipbuilding industry in recent years.**<sup>4</sup> Consequently, many firms are now sourcing their ships from China instead of South Korea. **The example suggests that a monetary policy shock in one partner country, as well as a combination of monetary policy shocks in a sparse set of partner countries, might be enough to affect the restructuring of a firm's supply chains.**

In this paper, we investigate whether and how Chinese firms adapt their supply chains in response to international monetary policy shocks. Existing studies have shown that financial constraints incentivize firms to participate in supply chains, especially forging ties with international trading partners (Minetti et al., 2019). Then, would firms tend to terminate suppliers when exposed to heightened monetary policy shocks transmitted from supply chains? Would firms establish new supply chain relationships to eliminate the adverse effects? Would firms switch to less exposed countries or re-shore their supply chains? Our study aims to examine these important yet unexplored questions and provide suggestive evidence on potential channels.

China provides an ideal experimental context to examine the effects of international monetary policy shocks rippling through trade networks for several reasons. First, China has strict capital account controls, partially ruling out the capital (portfolio) flows channel through which global monetary policy shocks affect the operational risk management of Chinese firms. This ensures that monetary policy shocks are primarily transmitted through trade networks. Second, China's status as the leading commodity trader and the prevalence of credit constraints among Chinese firms provide good insights into the nexus between financial frictions and trade, an essential question in international economics. Studies have shown that foreign ownership of Chinese firms significantly improves their financial conditions, and FDI firms are more likely to provide trade credit than local firms during contractionary monetary policy episodes (Jian & Wang, 2015; Lin & Ye, 2018). Then, we are strongly motivated to explore whether international trade networks, besides the FDI network, can propagate global monetary policy shocks to Chinese firms.

We combine detailed firm-to-firm level supply chain data from the Factset Revere with shipment data from the Panjiva provided by S&P Global, and financial statement information from the China Stock Market & Accounting Research Database (CSMAR). Our sample includes 1754 Chinese listed firms establishing relationships with 5726 unique suppliers across 78 countries for 2011–2018. We construct a firm-level measure of monetary policy shock, based on the supplier network in the preceding year and the exposure of suppliers to different monetary policy shocks across countries. Our analyses relate the supplier-side monetary policy shock with

<sup>4</sup> The shipbuilding industry is highly leveraged. Firms typically use shipbuilding orders as collateral to secure bank loans, then purchase raw materials like steel and equipment to carry out production. After delivering the ships and receiving payments, they repay the bank loans. The sharp rise in interest rates has significantly increased financing costs and eroded profits.

several supply chain variables in the estimations that control for a rich set of fixed effects, including firm, industry-year, and city-year, and several firm fundamental variables, particularly firm-level exposure to exchange rate shocks.

Adaptation in global supply chains can be decomposed into disruptions and formations. We first provide evidence that exposure to global monetary policy shocks (MPS henceforth) significantly increases the separation rate of Chinese firms. Specifically, a one-standard-deviation increase in a firm's exposure to MPS escalates the probability of separation with suppliers by 0.06, which amounts to about 0.131 standard deviations of the separation rate. The effects are persistent. Firms permanently terminate more suppliers when exposed to higher MPS. We then turn the focus to new entries. Our results show that firms exposed to higher MPS have lower entry rates, though insignificant. The patterns are similar no matter whether we define new entries as suppliers that temporarily broke the relationship and re-entered the supply chain network, or as suppliers that have never been touched in preceding years.<sup>5</sup> These findings are also robust across a range of alternative MPS measures, including long-term interest rates to capture the stance of long-term MPS, high-frequency MPS indicators, and changes in supplier-side MPS rather than levels.

We then take one step forward to separately examine how firms adjust their relationships with foreign and domestic suppliers in response to MPS. Firms might view uncertainties of MPS differently across domestic and foreign markets when making sourcing decisions. We find that the separation rates of both foreign and domestic suppliers are significantly and positively associated with the exposure of MPS. This implies that the overall MPS transmitted from trade networks detrimentally affects firms' normal operations and compels them to terminate suppliers regardless of their origins. Regarding new entries, we find that high exposure to MPS prevents firms from establishing new supply chain relationships in foreign markets while having negligible effects on domestic supply chains.

We find additional evidence of switching. Drawing on a firm-country-year balanced panel, we show that firms diversify among foreign suppliers in response to the risks brought by MPS. When the interest rate of a foreign country increases, firms are more likely to shift their suppliers from that country to other foreign countries but not return to China. We further confirm the importance of monetary conditions by showing that firms switch suppliers from high-interest-rate countries to low-interest-rate countries when exposed to severe global monetary policy shocks.

We notice that though supplier countries implement distinct monetary policies, U.S. monetary policy inevitably affects global financial conditions. To rule out the possibility that our results are exclusively driven by the U.S. side, we exclude all U.S. suppliers from our sample and find that the new measure of non-U.S. monetary policy shocks consistently influences the adaptation of Chinese firms' supply chains. We also show that MPS shocks from countries with either low or high correlation between their basic interest rates and the U.S. rate have a notable impact on supplier compositions. Our findings are also robust to alternative measures of supply chains and various subsamples.

To understand the underlying forces that drive the response to supplier-network-based MPS, we further examine how country-specific monetary policies affect the performance of Chinese firms' suppliers and identify the spillover effect. We confirm that contractionary monetary policies significantly increase interest expenses, which might deteriorate suppliers' funding conditions and increase production costs. Also, we provide an overall assessment of operating performance and show that tightening monetary conditions deteriorate supplier profitability. In light of the literature on the propagation of credit shocks to the real economy (Anna, 2020), we propose that supplier-side MPS negatively affects customers through two channels: trade credit and sales. We show suppliers facing tightening monetary conditions extend less trade credit to their customers. A one-standard-deviation increase in the basic interest rate decreases trade credit by 6.15% on average. The magnitude could be enough as an incentive for Chinese firms to adapt their supply chains. Furthermore, suppliers with unfavorable funding conditions may raise the price of their products or suppress production, leading to changes in total sales. We show that suppliers' sales remain flat but notably the price of products provided by supplier industries significantly hikes in response to an increase in basic rates. The exporting price for goods to China does not exhibit significant changes, probably because suppliers charging high prices have been terminated by Chinese customers, echoing the baseline patterns of separation rates.

The focus of this paper is the adaptation of Chinese customers. However, the disruption of each supply chain relationship can be driven by either the customer or the supplier decisions. On the one hand, Chinese firms may terminate a relationship when they perceive that the corresponding supplier has poor performance. On the other hand, a supplier negatively affected by poor funding conditions may take the initiative to break up the relationship with Chinese customers if it prioritizes the needs of other privileged customers. To disentangle the two forces, We adopt two sets of correlates to measure the relative strength of the customer and supplier: one is the availability of potential substitute partners, and the other is the bargaining power in terms of comparative advantage and competition. Turning to the disaggregated customer-supplier-year dataset, we find that the effect of MPS on the probability of disruption is more pronounced when Chinese customers have more potential suppliers or come from more specialized or less competitive industries. However, we find no evidence of the differential impacts based on the relative strength of suppliers. Admittedly, we cannot completely rule out the explanation that supplier decisions contribute to the disruption, but our preliminary evidence lends more support to a customer-driven story.

Next, we document the differential effects of MPS on supply chains based on firm and supply chain characteristics. Firms with a higher degree of external finance dependence and trade credit dependence, or from cities with fewer bank loan resources are more sensitive to input-share-weighted MPS. Long-duration customer-supplier relationships are less susceptible to disruption, while firms are more likely to terminate suppliers from geographically or institutionally distant countries. The heterogeneities imply that

<sup>5</sup> In a similar vein, Khanna et al. (2022) provides evidence that firms more exposed to external shocks from suppliers, that is, lockdown policies, have experienced significantly higher separation rates, while entry rates were not sensitive to the shock.

firms with stricter credit constraints, or supply chain relationships with less mutual trust or requiring higher trading costs, are more responsive to MPS.

Finally, we examine the real effects of MPS from suppliers. Regarding international trade activities, **we find that MPS substantially deteriorates trade performance.** To assess the overall impact on operating performance, we demonstrate that high exposure to MPS is associated with a significant decline in the growth rate of profits. **These findings provide suggestive evidence that even if firms are able to adjust their supply chain networks to counteract the MPS transmitted from trade networks, restructuring still imposes adverse effects on firm operations and international business.**

Our paper fits into the literature on the disruptions and resilience of supply chains from the perspective of monetary policy. The effects of external shocks and firms' responses, particularly diversification and reshoring, have been the focus. A few studies quantify the economic impacts and derive the equilibrium of the supplier network in response to idiosyncratic and aggregate shocks in theoretical models (Vasco et al., 2021; Matthew et al., 2022). Grossman et al. (2023) further conduct welfare-theoretic calculus and evaluate optimal policies to promote supply chain resilience. Other empirical studies leverage firm-to-firm supply chain data or bilateral trade data to examine the effects of different types of external shocks, such as natural disasters (Boehm et al., 2019; Balboni et al., 2024), climate change (M.C. & Christoph, 2024), economic uncertainties (Charoenwong et al., 2023), scandals (Pamina & Poncet, 2022), and terrorist attacks (Tan et al., 2024). **Our research differentiates from these studies by focusing on an unexplored fundamental shock, trade-network-based monetary policy shocks.** Supply chains can be an important channel of alleviating credit constraints for participants, and correspondingly, we provide evidence about how firms adjust their supply chain relationships when tightening monetary policies deteriorates the funding condition of their suppliers. We show that MPS leads Chinese firms to terminate more suppliers, establish fewer new supply chain relationships, and diversify among foreign suppliers.

Our paper also contributes to studies on the real effects and transmission of monetary policies. In a domestic setting, extensive literature has shown that monetary policies affect the real economy and examined specific channels like interest rate, credit, risk-taking, or investment channels (Ben & Gertler, 1995; Matteo et al., 2015; Ioannidou et al., 2015; Pablo & Winberry, 2020). Studies fairly emphasize firm heterogeneities in response to monetary policy shocks (Durante et al., 2022; E & Robert, 2018; Cloyne et al., 2023) which offer some insights for our analysis. In a global setting, studies mainly discuss the financial spillover effect of U.S. monetary policies on the rest of the world, particularly on emerging economies (Tatjana & Vasishtha, 2020; Silvia & Rey, 2020). Manuel et al. (2023) document that production networks are important to the transmission of monetary policy but focus on trade financing. Our work is the first to depict a firm's exposure to monetary policy shocks and examine its impacts on the restructuring of supply chains. As monetary policies largely determine credit conditions, this paper is also related to the propagation of financial shocks through input-output linkages and mechanisms including trade credit and price effects (Alfaro et al., 2021; Chen, 2023; Anna, 2020).

The remainder of this paper is structured as follows: Section 2 describes data and variables. Section 3 details the main empirical findings on the disruption, formation, and switching of supply chains. Section 4 discusses potential channels rationalizing empirical findings and provides preliminary evidence on disentangling customer- or supplier-driven forces. Section 5 details heterogeneity analyses. Section 6 discusses other real outcomes of monetary shocks transmitted from suppliers. Section 7 gives concluding remarks.

## 2. Data, variable definition, and sample construction

We examine the impact of trade-induced monetary policy spillover on supply chain stability, focusing on China as a unique and suitable setting. Our choice of China is driven by two primary reasons: First, China enforces strict capital controls and closely monitors cross-border capital flows, insulating Chinese firms from the volatility of global capital markets. This insulation minimizes the influence of cross-border financing on domestic firms, ensuring that monetary policy shocks are primarily transmitted through trading networks. Second, with its substantial involvement in international trade, China represents a significant portion of cross-border commodity transactions. **The combination of strict capital controls and active participation in international trade makes China an ideal context to isolate and examine the effects of cross-border monetary policy spillovers through trading networks.**

In this study, we employ three distinct datasets corresponding to our analytical framework: one dedicated to our primary independent variables, another for the dependent variable, and the third for control variables. The ensuing sections provide a detailed introduction to these datasets and elaborate on the definitions of the variables involved.

### 2.1. Monetary policy shock exposure: International trade data

Our primary focus in this study is the impact of international monetary policy shock spillover induced by trade. We acquire international trade data from the **Panjiva database provided by S&P Global.**<sup>6</sup> Panjiva furnishes extensive transaction-level information on international trades across multiple countries globally. We extract Chinese import data from this database. Each transaction record includes details such as the transaction date, the name of the Chinese importer, the source country, the HS code, and the value of imported goods. Subsequently, we cross-reference importer names with those of Chinese-listed firms to gauge the firm-level exposure to international monetary policy shocks transmitted through the trading network.

Specifically, the supplier-side international monetary policy shock exposure is constructed as follows:

$$SupplierMPS_{it} = \sum InputShare_{ik(t-1)} Rate_{kt} \quad (1)$$

<sup>6</sup> Panjiva dataset: <https://panjiva.com>.

where  $i$ ,  $k$ , and  $t$  stand for firm, country, and year, respectively.  $Rate_{kt}$  is the money market interest rate of country  $k$  in year  $t$ , obtained from the International Financial Statistics (IFS) provided by the International Monetary Fund (IMF).<sup>7</sup> As shown in Fig. 1, there is significant heterogeneity in the paths of money market interest rates across different countries, providing abundant variations for our analysis.

$InputShare_{ik(t-1)}$  represents the RMB share of inputs used by firm  $i$  in year  $t-1$  purchased from country  $k$  and is constructed as follows:

$$InputShare_{ik(t-1)} = \begin{cases} \frac{Import_{ik(t-1)}}{COGS_{i(t-1)}}, & \text{if } k \neq \text{China} \\ \frac{COGS_{i(t-1)} - \sum_{k \neq \text{China}} Import_{ik(t-1)}}{COGS_{i(t-1)}}, & \text{if } k = \text{China} \end{cases}$$

The set for country  $k$  encompasses the nations where the suppliers of firm  $i$  are based. For any foreign country  $k$ , we calculate the total value of imports by firm  $i$  from country  $k$  in the previous year ( $t-1$ ), utilizing the international transaction records. The  $InputShare_{ik(t-1)}$  represents the proportion of imports from country  $k$  relative to the total cost of goods sold. Conversely, if  $k$  refers to China, we determine the domestic input share as the remainder of the cost of goods sold after subtracting the total imports by firm  $i$  in year  $t-1$ , and then we normalize this value by the cost of goods sold. This approach allows us to investigate the effects of exposure to supplier-side monetary policy, applying a one-year lag to supply chain-related variables to mitigate concerns about reverse causality.

$SupplierMPS_{it}$ , as a firm-level measure for international money market interest rate exposure, thus captures how firm  $i$  is exposed to the monetary policy shocks of its suppliers.<sup>8</sup>

## 2.2. Supply chain adjustments

In this study, we investigate the impact of exposure to supplier-side monetary policy shocks on the stability of trade linkages with suppliers. We focus on variations in suppliers as our outcome variables. For the supply chain data, we use the FactSet Revere, which is widely recognized and employed in the literature (Michael & Wu, 2022; Manuel et al., 2023; Choi et al., 2024). FactSet Revere hand-collects and verifies supply chain data from diverse sources, including 10-K filings, conference call transcripts, and major news media reports (Senay et al., 2022), making it the most exhaustive resource for supply chain information currently available. We align this supply chain data with international trade records by retaining information for all Chinese-listed companies from 2010 to 2018.

As a preview, Fig. 2 presents the geographical distribution of suppliers for Chinese listed firms. We compile a list of all suppliers and count the number of suppliers from each country. The figure shows a broad distribution of suppliers around the world, with the top three foreign sourcing regions being the U.S., Japan, and the Euro area. The dispersed locations of suppliers provide rich details for examining potential changes in the supply chains of Chinese listed firms.

For our analysis, we calculate the annual total number of suppliers for each firm to track changes in supply chains. We introduce four key variables to effectively capture these dynamics. The first two variables focus on assessing the impacts of international monetary policy spillovers on existing suppliers. To this end, we specifically measure separation and termination rates, calculated as follows:

$$Separation\ Rate_{it} = \frac{\text{No. of supp. to } i \text{ in } t-1, \text{ who don't supply in } t}{[(\text{No. of supp. to } i \text{ in } t-1) + (\text{No. of supp. to } i \text{ in } t)]/2} \quad (2)$$

$$Termination\ Rate_{it} = \frac{\text{No. of supp. to } i \text{ in } t-1, \text{ who don't supply in any year thereafter}}{[(\text{No. of supp. to } i \text{ in } t-1) + (\text{No. of supp. to } i \text{ in } t)]/2} \quad (3)$$

$Separation\ Rate_{it}$  measures the separation rate of the suppliers for firm  $i$  in year  $t$ . The number of separated suppliers is defined as the number of year  $t-1$  suppliers of the firm that do not supply to  $i$  in year  $t$ . We scale it by the average number of suppliers of the firm in years  $t$  and  $t-1$ .  $Separation\ Rate_{it}$  captures how the pre-existing suppliers of the firms are affected in year  $t$ , whether temporarily or permanently.

To assess the persistence of the impacts driven by the international monetary policy spillover, we also calculate the number of terminated links with suppliers, measured by the number of year  $t-1$  suppliers of firm  $i$  that are absent in all of the following years starting from year  $t$ . The termination rate  $Termination\ Rate_{it}$  is the number of terminated suppliers scaled by the average number of suppliers.

The supplier-side monetary policy shock may not only affect the pre-existing linkages with suppliers, but also matter for the entry of new suppliers. To investigate this, we construct two measures on the entry of new suppliers:

$$Entry\ Rate_{it} = \frac{\text{No. of supp. to } i \text{ in } t, \text{ who didn't supply in } t-1}{[(\text{No. of supp. to } i \text{ in } t-1) + (\text{No. of supp. to } i \text{ in } t)]/2} \quad (4)$$

<sup>7</sup> Money market interest rate data in the International Financial Statistics: <https://data.imf.org/regular.aspx?key=63087881>.

<sup>8</sup> We also use an alternative version of long-run  $SupplierMPS$ . Considering that customer firms may respond differently to short-run and long-run MPS (if they could ex-ante identify the nature of the shocks), we replace the money market interest rate with 10-year government bond yield. Our results remain robust (Table A.1).



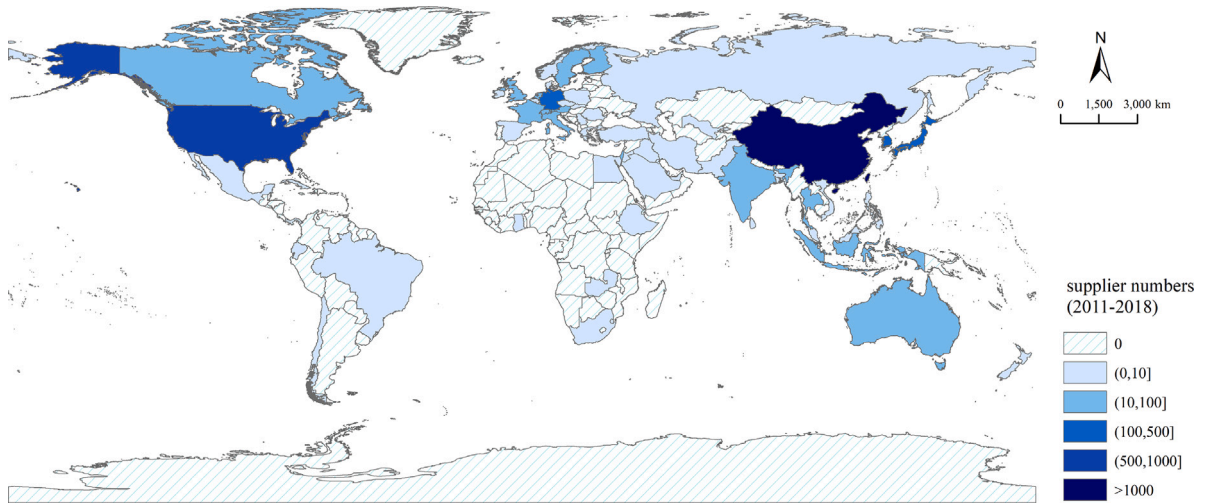


Fig. 2. The Geographical Distribution of Suppliers.

This figure illustrates the geographical distribution of suppliers for Chinese listed firms. We compiled a list of all suppliers, counted the number of suppliers from each country, and mapped them below. A deeper color indicates a larger number of suppliers.

$$Debut\ Rate_{it} = \frac{\text{No. of supp. to } i \text{ in } t, \text{ who didn't supply in any of the previous years}}{[(\text{No. of supp. to } i \text{ in } t-1) + (\text{No. of supp. to } i \text{ in } t)]/2} \quad (5)$$

$Entry\ Rate_{kt}$  measures the newly entered suppliers as a ratio of the average number of suppliers in two consecutive years. New suppliers are those that did not supply for firm  $i$  in year  $t-1$  but show up as firm  $i$ 's suppliers in year  $t$ . Note that under this definition, old suppliers with temporary disruption in year  $t-1$  would also be counted as new.

We also apply a stricter criterion, defining a supplier as new only if it has never been firm  $i$ 's supplier in any previous years. These suppliers are referred to as debut suppliers.  $Debut\ Rate_{it}$  then measures the ratio of debut suppliers over the average number of suppliers.

To further disentangle the sources of supplier changes, we divide the suppliers into two groups: foreign and Chinese suppliers. We then estimate entries and exits for the two groups separately. To be specific, the  $FSeparation\ Rate_{it}$  ( $CSeparation\ Rate_{it}$ ) is calculated as the number of separated foreign (Chinese) suppliers divided by the average number of suppliers.  $FTermination\ Rate_{it}$ ,  $CEntry\ Rate_{it}$ ,  $FDebut\ Rate_{it}$ , and  $CDebut\ Rate_{it}$  can be defined accordingly.

### 2.3. Other variables and summary statistics

We recognize that exchange rate fluctuations may also influence supply chain adaptations. Monetary policies not only affect the interest rate but also the foreign exchange rate which has sizable effects on exporting prices (Amiti et al., 2014; Tony, 2022). In our baseline regression, we add time fixed effects to account for potential exchange rate pass-through. Considering that the exposure to foreign exchange rate fluctuations depends on firms' trading networks, we additionally construct a firm-level real effective exchange rate ( $REER$ ) using a method similar to the construction of the supplier-side MPS. First, we obtain the real exchange rate for each supplying country from the International Financial Statistics (IFS). Then, for each firm in each year, we calculate the  $REER$  as the average real exchange rate weighted by the lagged input share. We include the exchange rate variable in all regressions.

We control for a set of firm characteristics, including total assets in log ( $Ln(Asset)$ ), debt to asset ratio ( $Leverage$ ), return on assets ( $ROA$ ), receivable ratio ( $Receivable$ ), revenue growth rate ( $RevGrowth$ ), net cash flow per share ( $CashFlow$ ), firm age ( $Ln(Age)$ ), and total holding ratio of top 5 shareholders ( $Top5Ratio$ ). The data are from the China Stock Market & Accounting Research (CSMAR) Database.

The final sample comprises 1754 Chinese listed firms observed from 2011 to 2018, totaling 11,910 observations. Table 1 presents the summary statistics. The average  $Separation\ Rate$  and  $Termination\ Rate$  stand at 0.159 and 0.153, respectively, indicating that supply chain disruptions and terminations are relatively common occurrences. Meanwhile, the average  $Entry\ Rate$  and  $Debut\ Rate$ , at 0.360 and 0.356, suggest frequent entries of new suppliers. These statistics collectively present a dynamic supply chain landscape characterized by continuous refreshment through both entries and exits. Understanding factors influencing these dynamics is pivotal for comprehending the evolution of firms' supply chains.

Regarding international monetary policy exposure, the average  $Supplier\ MPS$  stands at 2.657, with values ranging from a minimum of 0.559 to a maximum of 4.833. These statistics indicate a significant diversity in the monetary conditions encountered by suppliers (thus the firms), offering a robust foundation for our empirical investigation.

Table 1

Summary statistics.

This table presents the summary statistics. *Separation Rate<sub>it</sub>* is the number of suppliers to firm *i* in year *t* – 1 who do not supply to *i* in *t*, scaled by the average number of suppliers of *i* in *t* and *t* – 1. *Termination Rate<sub>it</sub>* is the number of suppliers to firm *i* in year *t* – 1 who do not supply to *i* any year afterward, scaled by the average number of suppliers of *i* in *t* and *t* – 1. *Entry Rate<sub>it</sub>* is the number of suppliers to firm *i* in year *t* who did not supply to *i* in *t* – 1, scaled by the average number of suppliers of *i* in *t* and *t* – 1. *Debut Rate<sub>it</sub>* is the number of suppliers to firm *i* in year *t* who did not supply to *i* in any of the previous years, scaled by the average number of suppliers of *i* in *t* and *t* – 1. *SupplierMPS* is the average money market interest rate of the suppliers, weighted by the input share. *REER* is the real effective exchange rate faced by each firm, calculated as the average real exchange rate with each supplier country weighted by the input share. *Ln(Asset)* and *Leverage* are the total asset (in log) and debt-to-asset ratio, respectively. *ROA* denotes the return on assets and *Receivable* is the sum of accounts receivable and bills receivable scaled by total assets. *RevGrowth* stands for the revenue growth rate and *CashFlow* is the net operating cash flow per share. *Ln(Age)* is the years of establishment (in log). *Top5Ratio* represents the aggregate holding ratio of the top 5 shareholders..

| Variable                | Obs    | Mean   | Std.Dev. | Min    | Max    |
|-------------------------|--------|--------|----------|--------|--------|
| <i>Separation Rate</i>  | 11,910 | 0.159  | 0.459    | 0.000  | 2.000  |
| <i>Termination Rate</i> | 11,910 | 0.153  | 0.449    | 0.000  | 2.000  |
| <i>Entry Rate</i>       | 11,910 | 0.360  | 0.688    | 0.000  | 2.000  |
| <i>Debut Rate</i>       | 11,910 | 0.356  | 0.685    | 0.000  | 2.000  |
| <i>SupplierMPS</i>      | 11,910 | 2.657  | 0.473    | 0.559  | 4.833  |
| <i>REER</i>             | 11,910 | 1.006  | 0.071    | 0.989  | 5.530  |
| <i>Ln(Asset)</i>        | 11,910 | 22.224 | 1.202    | 19.773 | 25.679 |
| <i>Leverage</i>         | 11,910 | 0.447  | 0.213    | 0.054  | 0.952  |
| <i>ROA</i>              | 11,910 | 0.038  | 0.061    | –0.232 | 0.214  |
| <i>Receivable</i>       | 11,910 | 0.120  | 0.103    | 0.000  | 0.462  |
| <i>RevGrowth</i>        | 11,910 | 0.183  | 0.45     | –0.542 | 2.966  |
| <i>CashFlow</i>         | 11,910 | 0.38   | 0.814    | –2.388 | 3.68   |
| <i>Ln(Age)</i>          | 11,910 | 2.832  | 0.324    | 1.792  | 3.434  |
| <i>Top5Ratio</i>        | 11,910 | 51.796 | 14.871   | 19.075 | 85.291 |

### 3. Empirical strategy and baseline findings

Monetary policy has important implications for real economic activities, and its transmission mechanisms have been extensively explored (Kashyap Anil & Jeremy, 2000; Mishkin, 2009; Blomhoff et al., 2021). This paper specifically investigates whether the supplier-side monetary policy shocks transmit along the supply chain, influencing the supplier choices and real performance of downstream firms.

#### 3.1. Empirical strategy

To examine how supplier-side monetary policy spillover through trading networks may contribute to changes in suppliers, we use a regression model as follows:

$$\begin{aligned} SuppChg_{it} = & \alpha_1 SupplierMPS_{it} + Controls + Firm_i \\ & + Industry_i \times Year_t + City_i \times Year_t + \epsilon_{it} \end{aligned} \quad (6)$$

*SuppChg<sub>it</sub>* represents the variables related to supplier changes, such as *Separation Rate* and *Entry Rate*. *SupplierMPS<sub>it</sub>* denotes the exposure of the firm to supplier-side monetary policy shocks, with its coefficient  $\alpha_1$  being the parameter of interest. To control for unobservable firm-specific attributes that are constant over time, we incorporate firm fixed effects. Additionally, to mitigate the influence of factors that vary over time at the industry and city levels, we control for industry  $\times$  year and city  $\times$  year fixed effects in all regressions.<sup>9</sup> Standard errors are clustered at the firm level.

Before diving into the empirical results, it is important to address potential identification challenges inherent in our study. Specifically, endogeneity concerns may arise from two main sources: reverse causality and omitted variable bias. Our independent variable, *SupplierMPS*, is derived from two distinct components: country-level interest rates and firm-country input shares. The likelihood that country-level interest rates are influenced by the actions of individual firms is minimal. Moreover, by utilizing input share data from the previous period, we diminish the possibility that these figures are affected by the firms' current supply chain decisions. Thus, the risk of reverse causality is not severe.

The omitted variables problem states that there could be latent factors simultaneously influencing the exposure to supplier-side monetary policy shocks and the restructuring of supply chains. Industry-specific shocks could be a potential confounding factor, as firms within the same industry often share similar supply chain dynamics and may be subject to comparable macroeconomic and

<sup>9</sup> We use the two-digit industries classified following the Guidelines for the Industry Classification of Listed Companies (2012 Revision) released by the Chinese Securities Regulatory Commission (CSRC). Cities are at the prefecture level.

**Table 2**

Supplier-side Monetary Policy Shocks (MPS) and changes in suppliers.

This table presents empirical results for the regression model as follows:  $SuppChg_{it} = \alpha_1 SupplierMPS_{it} + Controls + Firm_i + Industry_i \times Year_t + City_i \times Year_t + \epsilon_{it}$ . The dependent variables are the changes in suppliers, including *Separation Rate* (columns (1)–(2)), *Termination Rate* (columns (3)–(4)), *Entry Rate* (columns (5)–(6)), and *Debut Rate* (columns (7)–(8)). The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns.

|                           | <i>Separation rate</i> |                     | <i>Termination rate</i> |                     | <i>Entry rate</i> |                     | <i>Debut rate</i> |                     |
|---------------------------|------------------------|---------------------|-------------------------|---------------------|-------------------|---------------------|-------------------|---------------------|
|                           | (1)                    | (2)                 | (3)                     | (4)                 | (5)               | (6)                 | (7)               | (8)                 |
| <i>SupplierMPS</i>        | 0.127**<br>(0.052)     | 0.128**<br>(0.052)  | 0.129***<br>(0.050)     | 0.134***<br>(0.050) | −0.082<br>(0.077) | −0.085<br>(0.077)   | −0.075<br>(0.077) | −0.077<br>(0.077)   |
| <i>REER</i>               |                        | 0.010<br>(0.033)    |                         | 0.064**<br>(0.031)  |                   | −0.063<br>(0.054)   |                   | −0.064<br>(0.054)   |
| <i>Ln(Asset)</i>          |                        | 0.003<br>(0.013)    |                         | 0.002<br>(0.013)    |                   | 0.100***<br>(0.019) |                   | 0.101***<br>(0.019) |
| <i>Leverage</i>           |                        | −0.049<br>(0.055)   |                         | −0.058<br>(0.054)   |                   | 0.030<br>(0.082)    |                   | 0.026<br>(0.082)    |
| <i>ROA</i>                |                        | −0.106<br>(0.124)   |                         | −0.104<br>(0.123)   |                   | 0.028<br>(0.189)    |                   | 0.011<br>(0.189)    |
| <i>Receivable</i>         |                        | −0.029<br>(0.118)   |                         | −0.035<br>(0.116)   |                   | −0.089<br>(0.171)   |                   | −0.097<br>(0.172)   |
| <i>RevGrowth</i>          |                        | −0.026**<br>(0.011) |                         | −0.026**<br>(0.011) |                   | −0.018<br>(0.019)   |                   | −0.021<br>(0.019)   |
| <i>CashFlow</i>           |                        | 0.001<br>(0.009)    |                         | 0.001<br>(0.008)    |                   | 0.008<br>(0.011)    |                   | 0.007<br>(0.011)    |
| <i>Ln(Age)</i>            |                        | 0.032<br>(0.107)    |                         | 0.037<br>(0.106)    |                   | −0.267*<br>(0.149)  |                   | −0.277*<br>(0.149)  |
| <i>Top5Ratio</i>          |                        | −0.000<br>(0.001)   |                         | −0.000<br>(0.001)   |                   | −0.002**<br>(0.001) |                   | −0.002**<br>(0.001) |
| Firm FE                   | Yes                    | Yes                 | Yes                     | Yes                 | Yes               | Yes                 | Yes               | Yes                 |
| Industry $\times$ Year FE | Yes                    | Yes                 | Yes                     | Yes                 | Yes               | Yes                 | Yes               | Yes                 |
| City $\times$ Year FE     | Yes                    | Yes                 | Yes                     | Yes                 | Yes               | Yes                 | Yes               | Yes                 |
| Observations              | 11,910                 | 11,910              | 11,910                  | 11,910              | 11,910            | 11,910              | 11,910            | 11,910              |
| $R^2$                     | 0.331                  | 0.332               | 0.330                   | 0.331               | 0.317             | 0.319               | 0.316             | 0.318               |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

**sectoral disturbances.** These shared industry characteristics could lead to similar changes in supply chain arrangements, independent of monetary policy effects. Similarly, regional macroeconomic shocks could affect firms located in the same geographical area, leading to concurrent adjustments in their supply chain strategies. To mitigate these concerns, our analysis incorporates both industry  $\times$  year and city  $\times$  year fixed effects. By doing so, we account for unobserved heterogeneity and time-varying shocks at the industry and city levels, effectively controlling for potential omitted variable bias.

### 3.2. Supply chain adaptations and supplier-side MPS

This paper examines how exposure to supplier-side MPS drives supply chain adjustments. We begin with baseline results using *SupplierMPS* constructed from money market interest rates. To further mitigate potential confounding from socio-economic conditions embedded in annual interest rates and to better isolate monetary policy shocks, we supplement our analysis with alternative measures of MPS based on high-frequency data (Kenneth, 2001; Gurkaynak et al., 2005; Emi & Steinsson, 2018).

#### 3.2.1. Baseline results: Money market interest rate

Table 2 reports benchmark estimations with *SupplierMPS* as the key independent variable. Column (1) focuses on how supplier-side monetary policy shocks may contribute to the disruptions of supply chains. The significantly positive coefficient of *SupplierMPS* underscores the vulnerability of supply chains to supplier-side monetary policy shocks. A one standard deviation increase in the firm's exposure to supplier-side monetary policy shocks escalates the probability of separation with suppliers by 0.06, which amounts to about 0.131 standard deviations of *Separation Rate*.

Incorporating control variables in the analysis and observing a negligible change in the coefficient of *SupplierMPS* in column (2) reinforce the initial findings. The results in the first two columns suggest that the supplier-side monetary policy shocks can spill over through the trading network and have destabilizing effects on pre-existing supply chains. Firms exposed to tighter supplier-side monetary policy shocks are more likely to break up with their suppliers.

A natural follow-up question would be: is the negative impact transient or persistent? To see this, we focus on the termination rate in columns (3)–(4). The coefficients of *SupplierMPS* remain positive and significant at the 5% level. A one standard deviation increase in *SupplierMPS* raises the termination rate by 0.063, which is about 0.141 standard deviations of *Termination Rate*.



**Table 3**

Supplier-side MPS and changes in Suppliers - Foreign and Chinese suppliers.

This table explores the differential impacts of supplier-side monetary policy shocks on foreign and Chinese suppliers. The regression model is as follows:  $SuppChg_{it} = \alpha_1 SupplierMPS_{it} + Controls + Firm_i + Industry_i \times Year_t + City_i \times Year_t + \epsilon_{it}$ . Panel A focuses on supplier exits, with the dependent variables representing the separation and termination rates of foreign (*FSeparation Rate* and *FTermination Rate*) and Chinese suppliers (*CSeparation Rate* and *CTermination Rate*). Panel B instead investigates supplier exits and the dependent variables stand for the entry and debut rates of foreign (*FEntry Rate* and *FDebut Rate*) and Chinese suppliers (*CEntry Rate* and *CDebut Rate*). The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns.

| Panel A: Supplier Exits   |                         |                          |                         |                          |
|---------------------------|-------------------------|--------------------------|-------------------------|--------------------------|
|                           | <i>FSeparation rate</i> | <i>FTermination rate</i> | <i>CSeparation rate</i> | <i>CTermination rate</i> |
|                           | (1)                     | (2)                      | (3)                     | (4)                      |
| <i>SupplierMPS</i>        | 0.060**<br>(0.029)      | 0.058**<br>(0.028)       | 0.068*<br>(0.037)       | 0.076**<br>(0.036)       |
| Controls                  | Yes                     | Yes                      | Yes                     | Yes                      |
| Firm FE                   | Yes                     | Yes                      | Yes                     | Yes                      |
| Industry $\times$ Year FE | Yes                     | Yes                      | Yes                     | Yes                      |
| City $\times$ Year FE     | Yes                     | Yes                      | Yes                     | Yes                      |
| Observations              | 11,910                  | 11,910                   | 11,910                  | 11,910                   |
| <i>R</i> <sup>2</sup>     | 0.291                   | 0.291                    | 0.335                   | 0.331                    |
| Panel B: Supplier Entries |                         |                          |                         |                          |
|                           | <i>FEntry Rate</i>      | <i>FDebut Rate</i>       | <i>CEntry Rate</i>      | <i>CDebut Rate</i>       |
|                           | (5)                     | (6)                      | (7)                     | (8)                      |
| <i>SupplierMPS</i>        | -0.092*<br>(0.054)      | -0.090*<br>(0.054)       | 0.007<br>(0.054)        | 0.012<br>(0.054)         |
| Controls                  | Yes                     | Yes                      | Yes                     | Yes                      |
| Firm FE                   | Yes                     | Yes                      | Yes                     | Yes                      |
| Industry $\times$ Year FE | Yes                     | Yes                      | Yes                     | Yes                      |
| City $\times$ Year FE     | Yes                     | Yes                      | Yes                     | Yes                      |
| Observations              | 11,910                  | 11,910                   | 11,910                  | 11,910                   |
| <i>R</i> <sup>2</sup>     | 0.295                   | 0.293                    | 0.343                   | 0.341                    |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

Comparing the results in columns (1)–(2) and (3)–(4), we can see that the tightening in supplier-side monetary policies may exert permanent damage in existing trading networks, as witnessed by the termination of trading with old suppliers.

Shifting the focus from supply chain disruptions, columns (5)–(8) direct attention towards the entry of new suppliers. Columns (5)–(6) investigate how supplier-side monetary policy spillover influences the frequency of supplier entries. Despite being negative, the coefficients of *SupplierMPS* are statistically insignificant. **It is important to note that “supplier entries” in columns (5)–(6) encompass those that were absent in the previous year but exist in the current year.** This definition includes instances where old suppliers experienced temporary breaks in the prior year and are counted as new in the current year. To eliminate the influence of these temporary disruptions, columns (7)–(8) shift focus to the probability of debut suppliers. The findings in columns (7)–(8) mirror those in columns (5)–(6).

Our key independent variable, *SupplierMPS*, is constructed using money market interest rates to capture cross-country variations in monetary policy stances. However, the short-term nature of these rates may raise concerns about whether and why customers would respond to temporary monetary shocks. In addressing this concern, we note that although many monetary shocks may appear transitory in hindsight, firms face uncertainty about their persistence ex ante. This uncertainty motivates firms to respond to both short-term and long-term shocks. To further address this issue, we construct an alternative supplier-side MPS measure based on 10-year government bond yields, and our results remain robust (Table A.1).

**The empirical results indicate that stricter supplier-side monetary policies can spread through trading networks and lead to lasting disruptions in existing supply chains, with minimal effects on newly established connections.** To provide a more detailed analysis of supplier adjustments, we divide suppliers into two categories: foreign and Chinese suppliers. Subsequently, we examine the distinct effects of supplier-side monetary policy spillovers on these supplier groups. The findings are summarized in Table 3.

Panel A of Table 3 investigates the separation of firms from their existing foreign and Chinese suppliers. Columns (1)–(2) focus on the impact on foreign suppliers, echoing the trends observed in Table 2. The coefficients of *SupplierMPS* are positive and statistically significant, indicating that **heightened supplier-side monetary policy rates adversely affect existing trading networks.** A one standard deviation increase in *SupplierMPS* corresponds to a 0.027 rise in separation and termination rates among foreign suppliers. Similarly, columns (3)–(4) scrutinize Chinese suppliers, revealing analogous results. Here, the coefficients of *SupplierMPS* remain significantly positive, aligning closely with the magnitudes observed in columns (1)–(2). **This underscores that tightening monetary conditions for suppliers detrimentally impacts trading relationships, irrespective of supplier origin.**

Panel B of Table 3 analyzes the impact of supplier-side monetary policy shocks on the entries of new suppliers. Columns (1)–(2) examine the entry of new foreign suppliers, revealing that these shocks not only disrupt existing trading relationships but also hinder

the creation of new supply chains. A one standard deviation increase in supplier-side monetary policy shocks reduces the entry rate by 0.043. This effect is, however, insignificant among Chinese suppliers. **One plausible explanation for this discrepancy is that establishing foreign supply chains may necessitate more initial effort due to geographical and institutional differences. Consequently, the entry of new foreign suppliers is more susceptible to supplier-side monetary policy shocks.**

In summary, supplier-side monetary policy shocks can be transmitted through trading networks and disrupt existing supply chains. These disruptions affect both foreign and domestic suppliers to a similar extent. Additionally, the spillover leads to notable reductions in the entry of new foreign suppliers.

### 3.2.2. Results based on high frequency MPS

Our main independent variable, *SupplierMPS*, constructed as a weighted average of supplier-side money market interest rates, aims to capture the financing environment faced by a firm's suppliers. Nevertheless, using the level of interest rates may accidentally capture other socio-economic factors not necessarily related to monetary policy. While we show that our findings remain robust if we instead use the changes of supplier-side MPS (columns (3) and (4) of Table A.1), the changes may still be partially anticipated and therefore not constitute true "shocks". To address this concern, we provide additional results based on high-frequency monetary policy measures.

Specifically, these measures are constructed based on asset price movements within a narrow window around Federal Open Market Committee (FOMC) announcements and offer several advantages in our context: (i) The short window ensures that the captured price changes are attributable to monetary policy rather than confounding factors. (ii) By focusing on changes in federal funds futures prices, these measures net out market expectations prior to the FOMC and isolate the unexpected component of policy shocks. (iii) Because they use the principal component of multiple asset prices, including those affected by forward guidance and quantitative easing announcements, these high-frequency measures also capture the effects of unconventional monetary policy, partially alleviating concerns related to the zero lower bound.

Following the literature, we use three sets of high-frequency MPS measures: (i) *NSShock*: the measure proposed by Emi and Steinsson (2018); (ii) *KutterMPS*, based on Kenneth (2001); and (iii) the *target* and *path* factors from Gurkaynak et al. (2005). All three are constructed from changes in the pricing of federal funds futures or Eurodollar futures within a narrow window surrounding FOMC announcements and are widely used to identify unanticipated monetary policy shocks. A key limitation, however, is that these measures are primarily available only for U.S. monetary policy. As such, we present the corresponding results as a supplementary analysis.

We restrict the sample to Chinese firms with U.S. suppliers during our sample period and reconstruct it into a firm-year-quarter panel. The quarterly frequency is chosen to align with the timing of the high-frequency MPS measures. The dependent variables, aligned with those in baseline Eq. (6) include: (i) *US Separation Rate*, measured by the number of separated U.S. suppliers of customer firm  $i$  in quarter  $t$ , divided by the average number of U.S. suppliers of firm  $i$  in quarters  $t$  and  $t-1$ ; (ii) *US Termination Rate*, defined similarly using the number of terminated U.S. suppliers.

Table 4 presents the results based on high-frequency MPS. In line with our baseline results using annual money market interest rates, a higher high-frequency MPS is associated with a greater likelihood of supply chain disruptions.

### 3.3. Supplier switching

The previous analysis explores how the average supplier-side monetary policy shocks affect supply chain adjustments, with a particular focus on firm-level exit and entry dynamics. The findings suggest heightened monetary policy shocks raise the likelihood of breaking up with previous suppliers, possibly due to stricter borrowing constraints and increased production costs cascading through the supply chain. **Note, however, that the firm-level analysis fails to show explicitly that it is the tightening of monetary policy shock in one country, that leads to the separation of suppliers from that country.** The baseline results could be interpreted as increased monetary policy shocks in one place causing separations from other countries with minimal changes in money market interest rates. To clarify the origins of monetary policy shocks and their impacts on suppliers, we supplement with an additional firm-country level analysis.

We restructure our sample into a firm-country-year panel, creating a balanced dataset by retaining all supplier countries for each firm throughout the sample period. To ensure consistency and account for potential firm entries and exits, we only include firm-year records in the baseline sample. The revised regression model is as follows:

$$\begin{aligned} Supplier\_Change_{ikt} = & \beta_1 Rate_{kt} + Controls + Firm_i + Country_k \\ & + Industry_i \times Year_t + City_i \times Year_t + \epsilon_{ikt} \end{aligned} \quad (7)$$

where  $i$ ,  $k$ , and  $t$  denote firm, country, and year, respectively. **The main independent variable is  $Rate_{kt}$ , the money market interest rate of country  $k$  in year  $t$ .** The set of countries includes all supplier countries of firm  $k$  throughout the sample period, excluding China. The control variables are the same as those in the baseline analysis. We include firm, country, industry  $\times$  year, and city  $\times$  year fixed effects and cluster the standard error by firm.

We conduct three sets of analyses in this subsection. First, to capture how the monetary policy shock of country  $k$  affects the suppliers from the same country, we define *Separation Rate<sub>ikt</sub>* in Eq. (8) as the ratio of the number of separated suppliers of firm  $i$  from country  $k$  in year  $t$  over the average number of suppliers of firm  $i$  in year  $t$  and  $t-1$ . Separated suppliers are those that supply to firm  $i$  in year  $t-1$  but not in year  $t$ .

$$Separation\ Rate_{ikt} = \frac{No.\ of\ supp.\ from\ k\ to\ i\ in\ t-1,\ who\ don't\ supply\ in\ t}{[(No.\ of\ supp.\ to\ i\ in\ t-1) + (No.\ of\ supp.\ to\ i\ in\ t)]/2} \quad (8)$$

**Table 4**

High frequency MPS and changes in suppliers.

This table examines how supplier MPS affects supply chain adaptations based on high frequency MPS. We restrict the sample to Chinese customer firms with US suppliers during our sample period and reconstruct it into a firm-year-quarter panel. The dependent variable in Panel A is the *USSeparation Rate*, measured by the number of separated US suppliers of customer firm *i* in quarter *t*, divided by the average number of US suppliers of firm *i* in quarters *t* and *t* – 1; Panel B focuses on *USTermination Rate* instead, defined similarly using the number of terminated U.S. suppliers. We use three sets of high-frequency MPS measures: (i) *NSShock*: the measure proposed by Nakamura and Steinsson (2018); (ii) *KutterMPS*, based on Kuttner (2001); and (iii) the *Target* and *Path* factors from Gurkaynak et al. (2005). The Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

| Panel A.                                                  | <i>USSeparation Rate</i>  |                    |                    |                     |
|-----------------------------------------------------------|---------------------------|--------------------|--------------------|---------------------|
|                                                           | (1)                       | (2)                | (3)                | (4)                 |
| <i>NSShock</i>                                            | 0.700***<br>(0.222)       |                    |                    |                     |
| <i>KutterMPS</i>                                          |                           | 0.008**<br>(0.004) |                    |                     |
| <i>Target</i>                                             |                           |                    | 0.590*<br>(0.313)  |                     |
| <i>Path</i>                                               |                           |                    |                    | 0.143***<br>(0.044) |
| Controls                                                  | Yes                       | Yes                | Yes                | Yes                 |
| Firm FE, Industry $\times$ Year FE, City $\times$ Year FE | Yes                       | Yes                | Yes                | Yes                 |
| Observations                                              | 13,860                    | 13,860             | 13,860             | 13,860              |
| <i>R</i> <sup>2</sup>                                     | 0.118                     | 0.117              | 0.117              | 0.117               |
| Panel B.                                                  | <i>USTermination Rate</i> |                    |                    |                     |
|                                                           | (5)                       | (6)                | (7)                | (8)                 |
| <i>NSShock</i>                                            | 0.699***<br>(0.211)       |                    |                    |                     |
| <i>KutterMPS</i>                                          |                           | 0.009**<br>(0.004) |                    |                     |
| <i>Target</i>                                             |                           |                    | 0.619**<br>(0.297) |                     |
| <i>Path</i>                                               |                           |                    |                    | 0.141***<br>(0.042) |
| Controls                                                  | Yes                       | Yes                | Yes                | Yes                 |
| Firm FE, Industry $\times$ Year FE, City $\times$ Year FE | Yes                       | Yes                | Yes                | Yes                 |
| Observations                                              | 13,860                    | 13,860             | 13,860             | 13,860              |
| <i>R</i> <sup>2</sup>                                     | 0.119                     | 0.118              | 0.118              | 0.119               |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

Column (1) of Table 5, we present the results for the new measure. The coefficients of  $Rate_{kt}$  are significantly positive, suggesting that a higher money market interest rate in country *k* results in the separation of firms from suppliers in that country. This is in line with our baseline findings and provides more nuanced insights into how monetary policy shocks affect supplier composition.

Second, we examine how firms stabilize their sourcing strategies following the termination of suppliers from a specific country. As firms separate from suppliers affected by monetary policy shocks, it raises the question of how firms obtain the necessary inputs to sustain production. This motivates us to investigate the potential switching of suppliers induced by these shocks. For each supplier country *k*, we define  $CEntryRate_{ikt}$  and  $OtherFEntryRate_{ikt}$  as the changes in the numbers of suppliers from China and other foreign countries excluding *k*, as presented in Eqs. (9) and (10).

$$CEntryRate_{ikt} = \frac{\text{No. of supp. from China to } i \text{ in } t, \text{ who didn't supply in } t-1}{[(\text{No. of supp. to } i \text{ in } t-1) + (\text{No. of supp. to } i \text{ in } t)]/2} \quad (9)$$

$$OtherFEntryRate_{ikt} = \frac{\text{No. of supp. from foreign cnty. } (\neq k) \text{ to } i \text{ in } t, \text{ who didn't supply in } t-1}{[(\text{No. of supp. to } i \text{ in } t-1) + (\text{No. of supp. to } i \text{ in } t)]/2} \quad (10)$$

Columns (2)–(3) present regression results of switching. Column (2) focuses on the changes in the number of Chinese suppliers. The coefficient of  $Rate_{kt}$  is insignificant, indicating that increases in the money market interest rate of country *k* do not prompt firms to search for new Chinese suppliers. Instead, column (3) reveals a shift in suppliers to other foreign countries, with a significant rise in the entry rate of suppliers from foreign countries apart from *k*.

The firm-country level analysis above reveals that the tightening of monetary policy in one country can indeed lead firms to terminate relationships with suppliers there. Moreover, firms switch to other foreign countries to compensate for the loss of suppliers, as shown by the increased entry rate of suppliers in other foreign countries. This finding may seem to conflict with our findings on entry rate in Table 3, which suggests a decline in supplier entry rates. It is important to note that Table 3 consolidates data

**Table 5**

Supplier-side MPS and supplier switching of firms.

This table provides a more detailed examination of how supplier-side monetary policy shocks affect the supplier-switching behavior of the firms. This table perform a firm-country-level analysis:

$Supplier\_Change_{ikt} = \beta_1 Rate_{kt} + Controls + Firm_i + Country_k + Industry_i \times Year_t + City_i \times Year_t + \epsilon_{ikt}$  The main variable of interest is  $Rate_{kt}$ , the money market interest rate of country  $k$  in  $t$ .  $Separation\ Rate_{ikt}$  is the ratio of the number of separated suppliers of firm  $i$  from country  $k$  in year  $t$  over the average number of suppliers of firm  $i$  in year  $t$  and  $t - 1$ . For each supplier country  $k$ ,  $CEntry\ Rate_{ikt}$  and  $OtherFEntry\ Rate_{ikt}$  denote the changes in the numbers of new suppliers from China and foreign countries other than  $k$ , also scaled by the average number of suppliers of firm  $i$  in year  $t$  and  $t - 1$ . Columns (4)–(7) examines the directions of switching among suppliers and the four dependent variables (in log) are defined as follows:  $num\_lowrate$ , the number of suppliers from countries with interest rates lower than country  $k$ ;  $entrynum\_lowrate$ , the number of new suppliers from countries with interest rates lower than country  $k$ ;  $num\_highrate$ , the number of suppliers from countries with interest rates higher than country  $k$ ; and  $entrynum\_highrate$ , the number of new suppliers from countries with interest rates higher than country  $k$ . Control variables include total assets in log ( $Ln(Asset)$ ), debt to asset ratio ( $Leverage$ ), return on assets ( $ROA$ ), receivable ratio ( $Receivable$ ), revenue growth rate ( $RevGrowth$ ), net cash flow per share ( $CashFlow$ ), firm age ( $Ln(Age)$ ), and total holding ratio of top 5 shareholders ( $Top5Ratio$ ). To align with the data frequency, we control for the bilateral real exchange rate of country  $k$  in year  $t$ . We control for firm, country, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                           | <i>Separation rate</i> | <i>CEntry rate</i> | <i>OtherFEntry rate</i> | <i>Ln(num_lowrate)</i> | <i>Ln(entrynum_lowrate)</i> | <i>Ln(num_highrate)</i> | <i>Ln(entrynum_highrate)</i> |
|---------------------------|------------------------|--------------------|-------------------------|------------------------|-----------------------------|-------------------------|------------------------------|
|                           | (1)                    | (2)                | (3)                     | (4)                    | (5)                         | (6)                     | (7)                          |
| <i>Rate</i>               | 0.003*<br>(0.002)      | 0.0002<br>(0.005)  | 0.012**<br>(0.006)      | 0.033**<br>(0.017)     | 0.033***<br>(0.011)         | −0.066***<br>(0.020)    | −0.055***<br>(0.015)         |
| Controls                  | Yes                    | Yes                | Yes                     | Yes                    | Yes                         | Yes                     | Yes                          |
| Firm FE                   | Yes                    | Yes                | Yes                     | Yes                    | Yes                         | Yes                     | Yes                          |
| Country FE                | Yes                    | Yes                | Yes                     | Yes                    | Yes                         | Yes                     | Yes                          |
| Industry $\times$ Year FE | Yes                    | Yes                | Yes                     | Yes                    | Yes                         | Yes                     | Yes                          |
| City $\times$ Year FE     | Yes                    | Yes                | Yes                     | Yes                    | Yes                         | Yes                     | Yes                          |
| Observations              | 12,963                 | 12,963             | 12,963                  | 12,963                 | 12,963                      | 12,963                  | 12,963                       |
| $R^2$                     | 0.283                  | 0.499              | 0.438                   | 0.665                  | 0.544                       | 0.742                   | 0.653                        |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

across all supplier countries, reflecting an aggregate impact. This aggregation combines the negative impact on country  $k$  with the compensatory effect of switching to other foreign countries. Table 5, on the other hand, offers a more detailed breakdown of the effects of supplier-side monetary policy shocks, revealing a richer pattern by elucidating the dynamics of supplier switching across different countries.

Third, we examine the direction of supplier switching. While we have documented that firms switch to new suppliers in other foreign countries following an interest rate hike in a focal country, the evidence does not confirm an interest-rate-driven reallocation effect. If such reallocation were driven by money market rates, we would expect firms to shift towards countries with lower interest rates than the focal country and sever ties with those where rates are higher.<sup>10</sup>

To clarify this reallocation effect, we define four variables: (i)  $num\_lowrate$ , the number of suppliers from countries with interest rates lower than country  $k$ ; (ii)  $entrynum\_lowrate$ , the number of new suppliers from countries with interest rates lower than country  $k$ ; (iii)  $num\_highrate$ , the number of suppliers from countries with interest rates higher than country  $k$ ; and (iv)  $entrynum\_highrate$ , the number of new suppliers from countries with interest rates higher than country  $k$ .

Columns (4)–(7) present the results on supplier switching. We find that firms indeed shift sourcing from high-interest-rate to low-interest-rate countries. When interest rates rise in a given country, firms are more likely to retain or form relationships with suppliers in lower-rate countries (columns (4) and (5)), while reducing ties with suppliers in higher-rate countries (columns (6) and (7)). These results provide further insight into the directional pattern of supplier switching.

In summary, our analysis provides additional evidence that the tightening of monetary policy in a particular country leads to a termination of trading with suppliers from that country. Firms, in response, pivot towards suppliers in other low-rate foreign countries to diversify risks.

### 3.4. Robustness checks

#### 3.4.1. The spillover of U.S. monetary policies

We explore the impacts of supplier-side monetary policy shocks using the money market interest rates from countries where supplies are based. The implicit assumption is that suppliers in different countries face distinct monetary policy shocks. Nevertheless, many studies have documented the strong spillover effects of U.S. monetary policies on other regions, due to the economic and financial significance of the U.S. (Elias et al., 2019; Aeimit et al., 2021; Julian & Hale, 2022; Li et al., 2024). This raises a question regarding the interpretation of our findings: are our results driven by actual cross-country differences in monetary policies, or are they merely capturing the repercussions of U.S. monetary policies?

Fig. 1 presents the money market interest rates for the U.S. (the top foreign sourcing country) and other major supplying countries. From 2011 to 2018, the U.S. money market interest rate exhibited a clear upward trend. In contrast, the money market interest rates for Japan and the Euro area, the second and third largest foreign sourcing regions for Chinese listed firms, remained

<sup>10</sup> We thank an anonymous referee for this helpful suggestion

**Table 6**

Supplier-side MPS and changes in suppliers - U.S. MPS.

This table investigates to which extent the baseline findings are driven by U.S. monetary policy shocks. To save space, we only report the results on *Separation Rate* (odd columns) and *Termination Rate* (even columns). In columns (1)–(2), we exclude all U.S. suppliers and obtain a new supplier-side monetary policy measure *SupplierMPS\_NUS*, the average money market interest rate of non-U.S. suppliers, weighted by the input share. Considering the potential spillover of U.S. monetary policies to other countries, we also compute the correlation of money market interest rate with U.S. money market interest rate for each country. Countries with a U.S. correlation above 0.5 (median correlation: 0.489) are considered as highly correlated and the rest are grouped as those with low U.S. correlation. We then divide suppliers into two groups and compute supplier-side monetary policy shock measures for each group of suppliers. *SupplierMPS\_LowCorr* and *SupplierMPS\_HighCorr* indicate the supplier monetary policy shocks from suppliers located in countries with low and high U.S. correlation, respectively. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                             | Separation rate    | Termination rate   | Separation rate   | Termination rate   | Separation rate    | Termination rate   |
|-----------------------------|--------------------|--------------------|-------------------|--------------------|--------------------|--------------------|
|                             | (1)                | (2)                | (3)               | (4)                | (5)                | (6)                |
| <i>SupplierMPS_NUS</i>      | 0.127**<br>(0.061) | 0.138**<br>(0.059) |                   |                    |                    |                    |
| <i>SupplierMPS_LowCorr</i>  |                    |                    | 0.115*<br>(0.067) | 0.130**<br>(0.064) |                    |                    |
| <i>SupplierMPS_HighCorr</i> |                    |                    |                   |                    | 0.091**<br>(0.046) | 0.093**<br>(0.045) |
| Controls                    | Yes                | Yes                | Yes               | Yes                | Yes                | Yes                |
| Firm FE                     | Yes                | Yes                | Yes               | Yes                | Yes                | Yes                |
| Industry $\times$ Year FE   | Yes                | Yes                | Yes               | Yes                | Yes                | Yes                |
| City $\times$ Year FE       | Yes                | Yes                | Yes               | Yes                | Yes                | Yes                |
| Observations                | 11,699             | 11,699             | 11,623            | 11,623             | 10,452             | 10,452             |
| R <sup>2</sup>              | 0.337              | 0.335              | 0.338             | 0.337              | 0.357              | 0.357              |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

constant or declined over the same period. The divergence in cross-country money market interest rates suggests that our findings are less likely to be entirely driven by the spillover effects of U.S. monetary policies.

To further address this issue, we employ two approaches. First, we exclude all U.S. suppliers from our sample and reconstruct a non-U.S. supplier-side MPS measure. Columns (1)–(2) of Table 6 show that empirical results are robust to the exclusion of U.S. suppliers, indicating that our findings are not solely driven by U.S. suppliers.

Second, simply excluding U.S. suppliers does not eliminate the potential spillover effect of U.S. monetary policy on other countries. To investigate this, we calculate the correlation of each country's money market interest rate with the U.S. rate. Countries with a correlation above 0.5 are considered highly correlated, while the rest are classified as low-correlation countries. We then split the supplier-side monetary policy shock into two sub-measures based on the correlation: *SupplierMPS\_LowCorr* and *SupplierMPS\_HighCorr* represent the shocks from suppliers in low and high U.S. correlation countries, respectively. Columns (3)–(6) in Table 6 show that the monetary policy shocks from both groups affect supplier adaptations to a similar extent. These findings confirm that our results are driven by cross-country differences in monetary policies, rather than solely by the spillover effects of U.S. monetary policies.

### 3.4.2. Controlling for the impact of domestic MPS

The variable *SupplierMPS*, constructed as a weighted average of foreign and domestic interest rates, is designed to capture overall supplier-side funding shocks. However, combining both foreign and domestic rates may raise concerns regarding the interpretation of our results. For instance, a higher separation rate among a firm's foreign suppliers may reflect tighter monetary policy abroad or spillover effects from domestic (Chinese) monetary policy. While the inclusion of year fixed effects helps mitigate this issue, differences in domestic input shares imply heterogeneous exposure to domestic shocks. To further address this concern, we conduct two additional tests.

First, we construct an alternative measure of supplier-side MPS exposure using foreign suppliers only. Specifically, this measure is defined as follows:

$$SupplierMPS\_Foreign_{it} = \sum ImportShare_{ik(t-1)} Rate_{kt}$$

$$ImportShare_{ik(t-1)} = \frac{Import_{ik(t-1)}}{TotalImport_{i(t-1)}} \quad (11)$$

The key difference in this alternative measure is that, instead of input shares, we weight the foreign interest rate by the one-period lagged ratio of imports from country  $k$  to total imports. By excluding domestic interest rates from *SupplierMPS* and controlling for year fixed effects, we mitigate concerns about the direct impact of domestic monetary policy. Columns (1) and (4) of Table A.2 present the results with *SupplierMPS\_Foreign<sub>it</sub>* as the main independent variable, and the findings remain robust.

To further address potential confounding from the domestic policy environment and the composition of imported inputs, we include interaction terms between the share of imported inputs and year dummies. Columns (2)–(3) and columns (5)–(6) of Table A.2 present the results using *SupplierMPS* and *SupplierMPS\_Foreign* as key independent variables, respectively. The inclusion of these interaction terms does not alter our findings.



As an alternative test to disentangle foreign and domestic MPS effects, we turn to the firm-country-year panel used in the supplier-switching analysis in Section 3.3. The inclusion of the country-specific imported input share and year fixed effects absorbs variation in both imported input intensity and domestic MPS. As shown in Table A.3, firms are more likely to terminate supplier relationships in countries with high interest rates, while foreign MPS has limited impact on supplier entry. These results are consistent with our baseline finding that MPS transmitted through international trade networks disrupts supply chains.

In summary, we address potential confounding from domestic MPS by (i) constructing an alternative supplier-side MPS measure, (ii) controlling for interactions between input shares and year fixed effects, and (iii) leveraging a firm-country-year panel structure. Across all tests, our findings remain robust.

#### 3.4.3. Alternative measures of supply chain adjustment

The baseline analysis assesses supply chain disruptions or creations in percentage terms. In this subsection, instead of separately examining entries and exits, we conduct a direct test by analyzing the overall change in the number of suppliers. Column (1) in Table A.4 investigates the impact of supplier-side monetary policy shocks on the total number of suppliers (logged) in the current period. In line with the baseline findings, column (1) shows that heightened monetary policy shocks significantly reduce the number of suppliers. In column (2), we adopt the change in the number of suppliers and present a similar picture. Then, we explore the impact of supplier-side monetary policy shocks on the levels of supplier entries and exits. Columns (3) and (4) replace the dependent variables with the number of separated and terminated suppliers. Consistently, we observe increased supply chain disruptions and terminations with tightened supplier-side monetary policy shocks. Regarding new supplier entries, columns (5) and (6) suggest that supplier-side MPS do not significantly affect new supplier entries, aligning with the baseline findings.

#### 3.4.4. Using the customs data

Our baseline analysis uses the Panjiva dataset as it covers more up-to-date data than the Chinese Customs data, and this dataset has been employed in recent studies on international trade (Alfaro et al., 2025; Charoenwong et al., 2023), but its key limitation lies in its exclusive coverage of *seaborne trade*. In this subsection, we use the customs data to run analogous regressions and ensure our findings. Consistent with our baseline results, Table A.5 shows that exposure to heightened monetary policy shocks transmitted through the trade network significantly increases the separation and termination rates of Chinese firms.

#### 3.4.5. Excluding firms with certain characteristics

Though the whole sample demonstrates a clear negative impact of the supplier-side monetary policy shocks on existing supply chains, there might be concerns that the results are driven by a subset of firms whose supplier networks might be inherently vulnerable. For example, newly established firms might be in the initial stage of building up their supplier networks, during which process supplier adjustments might be frequent. If our sample is dominated by these firms, the findings may accidentally pick up these inherent changes and the results may not necessarily be linked to monetary policy shocks. We then exclude young firms established in less than 10 years to ensure robustness. Our main findings remain unchanged and the coefficients are close in magnitude to those of the baseline analysis.

In addition, how firms navigate supplier-side monetary policy shocks may be influenced by their standing in the market. Large firms, owing to their stronger bargaining power, improved access to credit, and market dominance, may handle external monetary policy shocks differently from smaller firms. To prevent bias from capturing the unique responses of the large-sized firms, we exclude firms ranking in the top decile by annual size. Table A.6 shows that removing the largest firms from our sample has minimal impact on the significance and magnitude of the coefficients.

In summary, our findings remain robust upon excluding firms with inherent instability or potentially different ways of responding to external shocks.

### 4. Why does MPS induce supply chain disruptions?

Upon establishing that supplier-side MPS results in significant supply chain adaptations, we delve deeper to understand the underlying economic forces. This involves a two-step analysis: First, we examine how MPS affects supplier performance, especially along dimensions critical to downstream customers. Second, we explore whether customer-side or supplier-side factors dominate the resulting supply chain disruptions.

#### 4.1. MPS and supplier performance

To examine the impacts of monetary policy changes on supplier performance, we compile a list of all suppliers for Chinese listed firms and match it with financial statement data from the Compustat database. As a preview, we show how monetary policy changes affect the overall performance of suppliers. Such changes significantly alter the funding conditions for suppliers. In line with this, columns (1) and (2) of Table 7 show that interest expenses, whether in log or as a ratio of total debt, significantly increase during periods of heightened monetary policy shocks. This implies that severe MPS makes it costly for firms to finance their investments and thereby, increases production costs. Suppliers cannot fully mitigate these unfavorable funding shocks, as evidenced by the negative responses in ROA shown in column (3). These results indicate that an increase in money market interest rates strongly deteriorates supplier performance.

Table 7

Monetary policy shocks and supplier performance.

This table examines how monetary policy changes affect supplier performance. Columns (1)–(6) perform the following regression using supplier-year data:  $Dep_{jkt} = \alpha_1 Rate_{kt} + Controls + Supplier_j + SIndustry_j \times Year_t + SupplyCountry_k + \epsilon_{jkt}$ . Supplier-year performance data is obtained from Compustat. The dependent variable  $Dep_{jkt}$  represents the year  $t$  performance of supplier  $j$  in the country  $k$ . Specifically, the dependent variables are the interest expenses (in log, column (1)), interest expenses over total debt (column (2)), ROA (column (3)), receivables (in log, column (4)), receivables over total sales (column (5)), and sales (in log, column (6)). The main variable of interest is  $Rate_{kt}$ , the money market interest rate of the supplying country  $k$  in year  $t$ . Control variables include the real exchange rate of the supplying country, total assets in log ( $Ln(Asset)$ ), debt to asset ratio ( $Leverage$ ), revenue growth rate ( $RevGrowth$ ), intangible asset ratio ( $Intangible$ ), and operating cash flow scaled by total asset ( $OCF$ ). We control for supplier, industry  $\times$  year, and country fixed effects in the first six columns. Columns (7)–(8) examine the price effects using the export price data from the Centre for Prospective Studies and International Information (CEPII). The regression model is:  $Ln(Price)_{kht} = \alpha_1 Rate_{kt} + Controls + HS6_h \times Year_t + HS6_h \times SupplyCountry_k + \epsilon_{kht}$ .  $Ln(Price)_{kht}$  is the average export price (in log) for 6-digit HS code products of country  $k$  in year  $t$ . Column (7) focuses on the overall export price while column (8) examines the export price to China. We control for the real exchange rate of the supplying country, HS code  $\times$  year, and HS code  $\times$  country fixed effects in columns (7)–(8).

|                                    | (1)                    | (2)                 | (3)                  | (4)                   | (5)                  | (6)               | (7)                 | (8)                |
|------------------------------------|------------------------|---------------------|----------------------|-----------------------|----------------------|-------------------|---------------------|--------------------|
|                                    | <i>Ln(InterestExp)</i> | <i>IntRatio</i>     | <i>ROA</i>           | <i>Ln(Receivable)</i> | <i>RecvRatio</i>     | <i>Ln(Sales)</i>  | <i>Ln(Price)</i>    | <i>Ln(CNPrice)</i> |
| <i>Rate</i>                        | 0.073***<br>(0.022)    | 0.001***<br>(0.000) | -0.931***<br>(0.237) | -0.024***<br>(0.009)  | -0.014***<br>(0.004) | -0.004<br>(0.007) | 0.014***<br>(0.002) | -0.004<br>(0.004)  |
| Controls                           | Yes                    | Yes                 | Yes                  | Yes                   | Yes                  | Yes               | No                  | No                 |
| Supplier FE                        | Yes                    | Yes                 | Yes                  | Yes                   | Yes                  | Yes               | No                  | No                 |
| Supplying Country FE               | Yes                    | Yes                 | Yes                  | Yes                   | Yes                  | Yes               | No                  | No                 |
| Supplier Industry $\times$ Year FE | Yes                    | Yes                 | Yes                  | Yes                   | Yes                  | Yes               | No                  | No                 |
| HS6 $\times$ Year FE               | No                     | No                  | No                   | No                    | No                   | No                | Yes                 | Yes                |
| HS6 $\times$ Supplying Country FE  | No                     | No                  | No                   | No                    | No                   | No                | Yes                 | Yes                |
| Observations                       | 15,908                 | 16,208              | 17,157               | 17,590                | 17,554               | 17,564            | 347,690             | 155,538            |
| $R^2$                              | 0.949                  | 0.722               | 0.874                | 0.985                 | 0.779                | 0.990             | 0.886               | 0.912              |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

We then explore how these adverse impacts affect downstream customers. Drawing from the production distortions discussed in Bigio and Jennifer (2016), we identify two main channels: (i) trade credit provision and (ii) sales. First, trade credit is an essential short-term funding source for firms and a key determinant in interfirm trade (Anna, 2020). Faced with elevated funding costs, suppliers may reduce trade credit provisions to their customers, thereby passing the credit shock along the supply chain and potentially destabilizing the production network. Column (4) of Table 7 shows that, suppliers reduce the amount of trade credit extended to their customers when the money market interest increases in their countries. Considering that trade credit is inherently related to total sales, we further isolate the trade credit channel by scaling receivables by sales in column (5). The negative impact on trade credit remains robust.<sup>11</sup>

In addition to the credit channel, the sales channel suggests that monetary policy shocks may affect suppliers' real performance by disrupting the production process. Higher funding costs may increase supplier's production expenses, leading to a decline in total output. Depending on their market power, suppliers may respond by raising prices, thereby passing some of the elevated costs along the supply chain (Chen, 2023). As sales are the product of total output and price, the overall impact of monetary policy shocks on total sales can be negative or insignificant. Column (6) presents the results on sales. The coefficient of interest rate is insignificantly negative.

To further investigate the sales channel, we examine how monetary policies affect suppliers' output prices using export price data from the Centre for Prospective Studies and International Information (CEPII). The 6-digit HS code level data documents export price changes between each export-import country pair. We restrict the sample to export-country-industry pairs that have supplied to Chinese listed firms. The disaggregated price data is collapsed into an export-country-HS-code average price, weighted by the 2010 nominal export flows to each destination country. We then analyze how monetary policy shocks in supplying countries affect their weighted average export price. After controlling for the real exchange rate, HS code  $\times$  year, and HS code  $\times$  supplier country fixed effects, we find that higher money market interest rates significantly increase the price of products provided by supplier industries, lending some support to the sales channel (see column (7)).

Furthermore, column (8) replaces the dependent variable with the export price to China. Interestingly, despite the increase in aggregate export prices, the export price to China is insignificantly related to the money market interest rate of the supplier country. This may be due to the switching pattern documented earlier. In response to supplier-side MPS, Chinese firms shift their sourcing from high-rate countries to low-rate countries, thereby avoiding an increase in realized import prices.<sup>12</sup>

<sup>11</sup> We recognize that, due to data limitations, we cannot confirm that trade credit extended to foreign buyers, particularly Chinese buyers, declines following a monetary tightening in the supplier country. To provide additional evidence, we draw on the Segment database from the Compustat, which provides some information on the geographical distribution of sales for North American firms. Firms are required to disclose geographical information only if they generate revenue or hold assets in more than one country and if the relevant customer accounts for at least 10% of total revenue or assets. By solely focusing on the subsample of Chinese firms' suppliers with substantial foreign sales recorded in this database, we find that a higher money market interest rate is associated with a significant decline in trade credit provision.

<sup>12</sup> The insignificant impact on the export price to China may be confusing at the first glance. A more detailed explanation is that when making purchasing decisions, firms compare the (would-be) prices offered by different suppliers, but the CEPII database only records the prices of completed transactions. Therefore, the insignificant impact on export prices to China may thus reflect the switching behavior we document.

To summarize, our findings show that suppliers experience significant adverse economic outcomes when exposed to monetary policy shocks. These negative impacts may further affect downstream customers through reduced trade credit provision and increased product prices.

#### 4.2. Supply chain disruptions: Customer or supplier choice?

Tightened monetary policies induce more stringent financial conditions for suppliers, which lead to reduced trade credit, increased product prices, and worse supplier performance. These factors, contributing to supply chain disruptions, can be potentially driven by decisions of either the customer or the supplier.

First, customers may voluntarily terminate supply chain relationships with MPS-affected suppliers. This suggests a customer-side narrative, where supply chain break-ups are more likely when customers have less difficulty in replacing unfavorable suppliers. Customers maintaining several supply chain relationships have greater flexibility to adjust to upstream shocks, facilitating supplier switching. Additionally, in industries where customers possess strong bargaining power, alternative suppliers are more readily available, increasing the likelihood of customer-driven supply chain disruptions. Thus, if customer force dominates, we conjecture that customers with more suppliers and greater bargaining power are more likely to sever relationships with affected suppliers when exposed to heightened MPS shocks.

Second, the supply chain disruptions could also be driven by supplier-side decisions, involving both within-supplier and across-supplier selection. The within-supplier selection implies that suppliers may adjust their trade credit provision and pricing strategy across their customer base, prioritizing the needs of privileged customers at the expense of others (Vicente, 2007; Guillermo, 2020). The across-supplier selection suggests that weaker suppliers are more likely to be held up by their customers and thus avoid risking the loss of any customers, absorbing all adverse impacts without passing them downstream (Grossman Sanford & Oliver, 1986). In contrast, better-positioned suppliers are more likely to pass the shock through supply chains, unconcerned about potential customer loss. Both within and across supplier selections are more likely when suppliers have a larger pool of customers and stronger bargaining power.

To examine whether customer-side or supplier-side narratives dominate in the MPS-induced supply chain disruptions, transaction-level supply chain data are required. We utilize the customer–supplier–year supply chains from the FactSet, and the regression model is as follows:

$$Exit_{ijt} = \alpha_1 SupplierMPS_{it} + Controls + Customer_i \times Supplier_j + CIndustry_i \times Year_t + SIndustry_j \times Year_t + City_i \times Year_t + \epsilon_{ijt} \quad (12)$$

where  $i$ ,  $j$ , and  $t$  represent Chinese customer firms, suppliers, and years, respectively. The dependent variable,  $Exit_{ijt}$ , is an indicator variable that equals one if the supply chain relationship between customer  $i$  and supplier  $j$  breaks in year  $t$ , and zero otherwise. The main independent variable is the supplier-driven MPS experienced by customer  $i$  in year  $t$ . Control variables include all customer-year variables used in the baseline analysis. A set of stringent fixed effects for customer–supplier pair, customer industry  $\times$  year, supplier industry  $\times$  year, and customer city  $\times$  year is also included.<sup>13</sup>

Drawing on the firm-to-firm supply chain data, we first confirm our baseline findings in column (1) in Panel A of Table 8. The results show that a higher supplier-side MPS is associated with a significantly higher probability of supply chain break-up. We then attribute the observed disruptions to customer-side and supplier-side factors.

The ability to adjust partly depends on the diversity of trading counterparties, which we proxy by the Herfindahl–Hirschman Index (HHI) based on the distribution of partners across countries. For each customer (supplier), we obtain the number of unique suppliers(customers) in each partner country and calculate the corresponding HHI. Firms with below-median HHI are considered highly diversified. Columns (2)–(3) of Panel A in Table 8 present the results for customers. As expected, supplier-side MPS induces more supply chain break-ups for firms with a more diversified supplier base.<sup>14</sup> However, columns (4)–(5) suggest that the impact of MPS on supply chain disruptions does not significantly differ for suppliers with different degrees of customer diversification.

Furthermore, the effect and manner in which supplier-side shocks influence customers hinge on the relative bargaining power between these entities. To capture the relative bargaining power, we use two measures:

- First, firms operating in highly competitive industries generally possess lower bargaining power due to the abundance of competitors (Dunlop John & Benjamin, 1942; Daniela & Klapper, 2016). To proxy for competition intensity, we utilize the industry-level Herfindahl index (HHI) for each country.<sup>15</sup> Industries with an HHI above the median are categorized as low competition, while those below are deemed to have high competition.

<sup>13</sup> We use SIC two-digit industries for the supplier industry.

<sup>14</sup> Though the coefficients for both groups are positive and significant, the between-group difference of coefficients is statistically significant. We run 1000 simulations and the empirical  $p$ -value is 0.092

<sup>15</sup> HHI index is calculated based on total assets. The HHI index for Chinese industries is computed based on the listed firm data obtained from the China Stock Market & Accounting Research (CSMAR) Database. The HHI indices for other supplying countries are calculated using listed firm data from the Compustat database.

**Table 8**

MPS-driven supply chain disruptions - The choice of customer or supplier?

This table explores whether supplier-side or customer-side choices lead to the MPS-induced supply chain disruptions using supply-chain level data. The regression model is as follows:  $Exit_{ijt} = \alpha_1 SupplierMPS_{jt} + Controls + Customer_i \times Supplier_j + C Industry_i \times Year_t + S Industry_j \times Year_t + City_i \times Year_t + \epsilon_{ijt}$ . The dependent variable  $Exit_{ijt}$  is a dummy variable that equals one if the supply chain relationship between Chinese customer firm  $i$  and supplier firm  $j$  stops in year  $t$ , and zero otherwise. The main independent variable is the  $SupplierMPS$ , the weighted average supplier-induced money market interest rate. Panel A presents the whole sample results, together with the subsample analysis on diversity. Customer (supplier)-side diversity is measured by the Herfindahl–Hirschman Index (HHI) based on the supplier (customer) countries. Firms with below-median HHI are considered as highly-diversified. Panel B instead presents the subsample analysis on bargaining power, proxied by the Herfindale Index (HHI) and industry specialization. To avoid capturing domestic industry structural variations, we base our HHI and SP measures on 2010 data. For all subsample analyses, the sample is divided by comparing the grouping variable to the median. We control for customer firm-level control variables including real effective exchange rate ( $REER$ ), total assets in log ( $Ln(Asset)$ ), debt to asset ratio ( $Leverage$ ), return on assets ( $ROA$ ), receivable ratio ( $Receivable$ ), revenue growth rate ( $RevGrowth$ ), net cash flow per share ( $CashFlow$ ), firm age ( $Ln(Age)$ ), and total holding ratio of top 5 shareholders ( $Top5Ratio$ ). Customer  $\times$  supplier pair, customer industry  $\times$  year, supplier industry  $\times$  year, and customer city  $\times$  year fixed effects are included in all columns..

| Panel A. Whole sample results and subsample analysis based on diversity |                    |                                      |                     |                                      |                    |
|-------------------------------------------------------------------------|--------------------|--------------------------------------|---------------------|--------------------------------------|--------------------|
|                                                                         | Whole sample       | Customer side - Supplier country HHI |                     | Supplier side - Customer country HHI |                    |
|                                                                         |                    | High                                 | Low                 | High                                 | Low                |
|                                                                         | (1)                | (2)                                  | (3)                 | (4)                                  | (5)                |
| <i>SupplierMPS</i>                                                      | 0.025**<br>(0.010) | 0.035*<br>(0.020)                    | 0.068***<br>(0.026) | 0.030*<br>(0.016)                    | 0.028**<br>(0.013) |
| Controls, City $\times$ Year FE                                         | Yes                | Yes                                  | Yes                 | Yes                                  | Yes                |
| Customer Industry $\times$ Year FE                                      | Yes                | Yes                                  | Yes                 | Yes                                  | Yes                |
| Supplier Industry $\times$ Year FE                                      | Yes                | Yes                                  | Yes                 | Yes                                  | Yes                |
| Customer $\times$ Supplier FE                                           | Yes                | Yes                                  | Yes                 | Yes                                  | Yes                |
| Observations                                                            | 47,124             | 22,793                               | 23,442              | 22,682                               | 22,454             |
| $R^2$                                                                   | 0.229              | 0.267                                | 0.264               | 0.264                                | 0.266              |

| Panel B. Subsample analysis based on bargaining power |                        |                   |                    |                   |                         |                    |                   |                    |
|-------------------------------------------------------|------------------------|-------------------|--------------------|-------------------|-------------------------|--------------------|-------------------|--------------------|
|                                                       | Industry concentration |                   |                    |                   | Industry specialization |                    |                   |                    |
|                                                       | Customer-side          |                   | Supplier-side      |                   | Customer-side           |                    | Supplier-side     |                    |
|                                                       | Low<br>(6)             | High<br>(7)       | Low<br>(8)         | High<br>(9)       | Low<br>(10)             | High<br>(11)       | Low<br>(12)       | High<br>(13)       |
| <i>SupplierMPS</i>                                    | 0.017<br>(0.012)       | 0.062*<br>(0.034) | 0.034**<br>(0.016) | 0.029*<br>(0.015) | 0.013<br>(0.020)        | 0.031**<br>(0.014) | 0.024*<br>(0.013) | 0.030**<br>(0.015) |
| Controls, City $\times$ Year FE                       | Yes                    | Yes               | Yes                | Yes               | Yes                     | Yes                | Yes               | Yes                |
| Customer Industry $\times$ Year FE                    | Yes                    | Yes               | Yes                | Yes               | Yes                     | Yes                | Yes               | Yes                |
| Supplier Industry $\times$ Year FE                    | Yes                    | Yes               | Yes                | Yes               | Yes                     | Yes                | Yes               | Yes                |
| Customer $\times$ Supplier FE                         | Yes                    | Yes               | Yes                | Yes               | Yes                     | Yes                | Yes               | Yes                |
| Observations                                          | 24,831                 | 21,174            | 25,113             | 21,651            | 18,638                  | 27,533             | 22,859            | 23,224             |
| $R^2$                                                 | 0.253                  | 0.278             | 0.258              | 0.270             | 0.284                   | 0.238              | 0.268             | 0.257              |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at customer firm level.

- Second, product specialty grants firms unique industry-specific intangible advantages, placing them in more privileged conditions when bargaining with trading partners (Dass et al., 2015; Laurent et al., 2017). Following Laurent et al. (2017), we compute the industry specialization (SP) measure as the ratio of each industry's production in China or the supplier country to the average global proportion for that industry. Firms operating in industries with SP values above the median are deemed more specialized, thereby possessing stronger bargaining power.

Panel B presents the subsample analysis from the customer and supplier perspectives. On the customer side, we observe that customers in less competitive or highly specialized industries are more likely to sever ties with their suppliers during periods of heightened supplier-side MPS. This supports the customer-side narrative, as break-ups are more frequent when customers have a high comparative advantage and alternative suppliers are readily available. On the supplier side, however, supply chain break-ups do not significantly differ across different groups of suppliers. This suggests that the supplier-side narrative is less influential in our setting.

Taken together, we delve deeper into the economic mechanisms driving supply chain disruptions from both customer and supplier perspectives. While we cannot completely rule out the influence of supplier decisions, the findings align more closely with the customer-side narrative.

## 5. When does supplier MPS matter more?

The findings above indicate that firms choose to sever ties with their current suppliers in response to increased supplier-side MPS, due to the worsened credit provision and heightened commodity prices. To enrich our findings, we investigate key factors influencing

transactions and, consequently, the establishment and resilience of supply chains. We delineate three potential dimensions that matter: financial constraints, relationship strength, and transaction costs.

### 5.1. Stringent financial constraint

The supply chain entails both commodity trading and financing relationships. As noted in A and Raghuram (1997), trade credit plays a significant and extensively studied role in interfirm trade, serving as a crucial source of short-term external funding for companies. The previous literature has outlined how credit shocks may propagate along the supply chain (Manova, 2013; Anna, 2020; Manuel et al., 2023). Our previous results suggest that suppliers significantly reduce credit provision to their customers when exposed to tightened monetary policies. These shifts in trade credit provisions can ultimately contribute to supply chain disruptions and terminations.

Customers experiencing stricter financial constraints are expected to be more responsive to shifts in suppliers' financial conditions. To see this, we use three proxies for the tightness of financial constraints. The first measure is the industry-level external finance dependence, defined as the share of capital expenditures not covered by internal cash flows. This measure was first introduced in Raghuram and Zingales (1998) and has since been widely used in the literature (Cetorelli & Gambera, 2001; Viral & Xu, 2017).

The second measure we consider specifically focuses on the financing role of supply chains. Following Kalina et al. (2015), we calculate the ratio of changes in accounts payable to changes in total assets to derive an industry-level measure of trade credit dependence. For the first two measures, we compute the industry financial constraint measures using U.S. data spanning from 1980 to 1999. We then employ these U.S.-derived metrics as proxies for the financial constraints encountered by Chinese firms. This is the common practice in the literature for reasons including using the U.S. as a reference country to exclude the impacts driven by domestic financial development (Kalina et al., 2015). The rank-preserving nature of financial constraint measures alleviates concerns about the potential differences in absolute levels.

Third, we also consider the regional heterogeneity in the abundance of financial resources. We obtain the data on the annual outstanding volume of bank loans for each prefecture city. Firms located in cities with an outstanding bank loan volume in the last year above the median are considered those with low financial constraints, while the rest of the firms are those with high financial constraints.

Table 9 presents the results for firms grouped by the tightness of financial constraints. Panel A decomposes the firms into two groups depending on industry external finance dependence. Columns (1)–(2) and (3)–(4) include results for firms with low and high external finance dependence. As expected, the impact of supplier-side monetary policy spillover on supply chains is more pronounced among firms facing tight financial constraints. Panels B and C apply the other two financial constraint measures, yielding results consistent with those in Panel A.

Taken together, our analysis reveals that supplier-side monetary policy shocks can disrupt supply chains by exacerbating financial constraints. Notably, our empirical findings indicate that firms with tighter financial constraints experience more pronounced effects.

### 5.2. Weak firm-supplier relationship

In addition to the exchange of goods and money, frequent trading between firms and suppliers also facilitates the exchange of information, laying the groundwork for the development of relationships (Gerard et al., 2000). These trading relationships have significant implications for the economic activities between customers and suppliers. The probability of trading is higher for firm-supplier pairs with established relationships. Moreover, suppliers are more inclined to offer price discounts and other trading-related benefits, such as trade credit, to their relational customers (A & Raghuram, 1997). Building on the relational trading literature, we expect that the strength of the trading relationship also matters for the impacts of supplier-side monetary policy shocks on supplier composition.

Stronger firm-supplier relationships are more resilient to external shocks as firms are incentivized to maintain long-term relational benefits. To empirically examine how the impacts of supplier-side monetary policy shocks vary depending on the strength of supplier relationships, we use the relationship duration as a proxy. Generally, relationships strengthen as their duration increases. For each firm-supplier pair in year  $t$ , the relationship duration is the number of times the supplier supplied to the firm from 2003 to year  $t - 1$ .<sup>16</sup> For a firm in year  $t$ , we define long(short)-duration suppliers as those who supply to the firm in year  $t - 1$  with a duration above (below) the median among all year  $t - 1$  suppliers. We then separately calculate the separation and termination rates for these two groups of suppliers.

Columns (1)–(2) and (3)–(4) of Table 10 report the results on short-duration and long-duration suppliers, respectively. In line with our expectations, relationships with longer duration are less vulnerable to exogenous shocks. Among the short-duration suppliers, both separation and termination rates are significantly positive, indicating an increased number of broken supply chains. In contrast, for long-duration suppliers, supplier-side monetary policy shocks have negligible impacts on supplier composition.

<sup>16</sup> The starting year is set at 2003 due to the limitations of the FactSet database.



**Table 9**

Supplier-side MPS and changes in suppliers - Financial constraint.

This table explores how the impact of supplier-side monetary policy shocks on supplier composition differs depending on the financial constraints of the firm. Panel A, B, and C use external finance dependence, dependent on trade credit, and local abundance of financial resources as proxies for the tightness of financial constraints, respectively. For all subsample analyses, the sample is divided by comparing the grouping variable to the yearly median. To save space, we only report the results on *Separation Rate* (odd columns) and *Termination Rate* (even columns). The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns.

|                                                    | Low financial constraint |                         | High financial constraint |                         |
|----------------------------------------------------|--------------------------|-------------------------|---------------------------|-------------------------|
| Panel A: External finance dependence               |                          |                         |                           |                         |
|                                                    | <i>Separation rate</i>   | <i>Termination rate</i> | <i>Separation rate</i>    | <i>Termination rate</i> |
|                                                    | (1)                      | (2)                     | (3)                       | (4)                     |
| <i>SupplierMPS</i>                                 | 0.081                    | 0.086                   | 0.182*                    | 0.182*                  |
|                                                    | (0.069)                  | (0.064)                 | (0.096)                   | (0.096)                 |
| Controls                                           | Yes                      | Yes                     | Yes                       | Yes                     |
| Firm FE                                            | Yes                      | Yes                     | Yes                       | Yes                     |
| Industry $\times$ Year FE                          | Yes                      | Yes                     | Yes                       | Yes                     |
| City $\times$ Year FE                              | Yes                      | Yes                     | Yes                       | Yes                     |
| Observations                                       | 5,815                    | 5,815                   | 5,217                     | 5,217                   |
| $R^2$                                              | 0.375                    | 0.374                   | 0.364                     | 0.364                   |
| Panel B: Dependence on Trade Credit                |                          |                         |                           |                         |
|                                                    | (5)                      | (6)                     | (7)                       | (8)                     |
| <i>SupplierMPS</i>                                 | 0.117                    | 0.117                   | 0.148**                   | 0.148**                 |
|                                                    | (0.095)                  | (0.095)                 | (0.071)                   | (0.066)                 |
| Controls                                           | Yes                      | Yes                     | Yes                       | Yes                     |
| Firm FE                                            | Yes                      | Yes                     | Yes                       | Yes                     |
| Industry $\times$ Year FE                          | Yes                      | Yes                     | Yes                       | Yes                     |
| City $\times$ Year FE                              | Yes                      | Yes                     | Yes                       | Yes                     |
| Observations                                       | 4,333                    | 4,333                   | 6,675                     | 6,675                   |
| $R^2$                                              | 0.380                    | 0.380                   | 0.360                     | 0.360                   |
| Panel C: Abundance of Regional Financial Resources |                          |                         |                           |                         |
|                                                    | (9)                      | (10)                    | (11)                      | (12)                    |
| <i>SupplierMPS</i>                                 | 0.128                    | 0.125                   | 0.196***                  | 0.207***                |
|                                                    | (0.092)                  | (0.090)                 | (0.072)                   | (0.071)                 |
| Controls                                           | Yes                      | Yes                     | Yes                       | Yes                     |
| Firm FE                                            | Yes                      | Yes                     | Yes                       | Yes                     |
| Industry $\times$ Year FE                          | Yes                      | Yes                     | Yes                       | Yes                     |
| City $\times$ Year FE                              | Yes                      | Yes                     | Yes                       | Yes                     |
| Observations                                       | 5,446                    | 5,446                   | 5,582                     | 5,582                   |
| $R^2$                                              | 0.289                    | 0.288                   | 0.418                     | 0.418                   |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

### 5.3. High trade cost

The literature on international trade has emphasized the trade cost being an indispensable component of comparable advantage, perhaps even more important than production costs (Obstfeld & Rogoff, 2000; James & Eric, 2003). Among others, distance stands out as a well-known determinant of trade costs, prominently featured in the classic gravity model (Matias & Freund, 2008; Krauthheim, 2012; Baier et al., 2018; Thomas, 2018). Apart from geographic distance, cultural and institutional differences also contribute to the overall distance between suppliers and customers (Porta et al., 2008). Greater distance between suppliers and customers increases trade costs, reduces the probability of trade, and heightens supply chain instability.

The supplier-side monetary policy shocks create harsher financing conditions for the suppliers and push up production costs, leading to heightened product prices. The increased production costs and higher product prices threaten the sustainability of the supplier-customer relationship, especially for long-distance links inherently more fragile. Consequently, we anticipate a heightened likelihood of separation between firms and suppliers located farther apart, reflecting the increased vulnerability of such relationships to these shocks.

To capture the distance between the firm and the supplier country, we use three measures sourced from the Centre for Prospective Studies and International Information (CEPII): First, we obtain the data for the bilateral distance between countries. The geographic distance between two countries is measured as the population-weighted distance between their most populated cities, using arithmetic mean. The distance is considered zero for domestic suppliers. Second, we measure the distance between suppliers and firms depending on whether the countries they are located in are contiguous. Contiguous countries are considered close. The third distance measure concerns the institutional distances and measures whether the supplier country shares the same

**Table 10**

Supplier-side MPS and changes in suppliers - Relationship strength.

This table investigates the differential impacts of supplier-side monetary policy shocks on supplier composition for customer-supplier relationships that are of different strength. We use the duration of the firm-supplier relationship as a proxy for relationship strength. For a firm-supplier pair in year  $t$ , duration is measured as the number of times that the supplier supplied to the firm from 2003 to year  $t - 1$ . For a firm in year  $t$ , long-duration suppliers are those who supply to the firm in year  $t - 1$  and with duration above the median among all year  $t - 1$  suppliers. To save space, we only report the results on *Separation Rate* (odd columns) and *Termination Rate* (even columns). The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log ( $\ln(\text{Asset})$ ), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age ( $\ln(\text{Age})$ ), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns.

|                           | Short duration         |                         | Long duration          |                         |
|---------------------------|------------------------|-------------------------|------------------------|-------------------------|
|                           | <i>Separation rate</i> | <i>Termination rate</i> | <i>Separation rate</i> | <i>Termination rate</i> |
|                           | (1)                    | (2)                     | (3)                    | (4)                     |
| <i>SupplierMPS</i>        | 0.128***<br>(0.049)    | 0.131***<br>(0.048)     | -0.0001<br>(0.011)     | 0.003<br>(0.011)        |
| Controls                  | Yes                    | Yes                     | Yes                    | Yes                     |
| Firm FE                   | Yes                    | Yes                     | Yes                    | Yes                     |
| Industry $\times$ Year FE | Yes                    | Yes                     | Yes                    | Yes                     |
| City $\times$ Year FE     | Yes                    | Yes                     | Yes                    | Yes                     |
| Observations              | 11,910                 | 11,910                  | 11,910                 | 11,910                  |
| $R^2$                     | 0.319                  | 0.318                   | 0.325                  | 0.322                   |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

legal origin as China. Countries sharing the same legal origin are considered closer in distance. To avoid capturing the time series changes in supplier composition, we obtain the average distance between firms and their suppliers based on the 2011 data.

Table 11 explores how trade costs contribute to supply chain instability. Panel A, B, and C depict the findings on firms with varying distances with their suppliers based on the three distance measures. The impact of supplier-side monetary policy shocks on the separation of suppliers mainly concentrates among firms situated farther away from their suppliers. This aligns with the notion that longer distances inherently increase supply chain vulnerability.

## 6. Further discussion on the real economic outcomes

Monetary policy shocks generate supply chain instability, but would they affect real economic outcomes? Table 7 shows that monetary policy shocks contribute to adverse economic outcomes for suppliers, and we are interested in whether the supplier-side shocks propagate along the supply chain, leading to negative economic outcomes for downstream firms. It has been documented that supply chain disruptions lead to long-term negative stock performance (B & Vinod, 2005). But if firms could mitigate the negative impacts by quickly spotting a suitable supplier as a substitute, the adverse impacts might be saved. In such a scenario where there is frictionless switching of suppliers from high-rate to low-rate countries, the monetary policy shocks may not affect firm performance, which makes them less of a concern.

To explore how supplier-side monetary policy shocks affect firms' real outcomes, we first examine the direct impacts on firm import. Echoing the baseline findings that MPS drives more foreign supply chain disruptions, Column (1) of Table 12 shows that the volume of imports significantly reduced when firms are exposed to higher supplier-side monetary policy shocks. In addition to the decline in import value, column (2) indicates that import diversity shrinks as supplier-side monetary policy tightens.

The negative impacts on the import side raise questions about the potential influence on the export side. Several studies document that the use of imported intermediates plays a role in determining product quality and overall productivity (Hiroiyuki & Rodrigue, 2008; Kalina & Yu, 2017). We then explore the potential impacts on firm export behaviors. Columns (3) and (4) show that supplier-induced monetary policy shocks also lead to significant declines in total export value and export quantity. More troubling is the considerable decline in export quality, as shown in column (5). This decrease in export quality aligns with the idea that supplier-side monetary policy shocks disrupt the production process, thereby damaging product quality. The results on exports also indicate that these import-side shocks could potentially extend beyond the firm, influencing the firm's customers.

To further shed light on changes on the export side, we distinguish between exports to developed and developing countries. Conceptually, selling to developed countries induces firms to purchase higher-quality inputs and upgrade the average quality of goods they produce (Paulo et al., 2018). When the adjustment of supply chains, such as matching and searching frictions, raises the costs of imported inputs or restricts access to critical foreign intermediates, firms' ability to produce high-quality products at low costs is significantly impaired. However, within a given firm-product, firms can set higher prices in richer countries and then earn bigger revenues (Kalina & Zhang, 2012). It is also plausible that firms reallocate scarce goods, that is, sacrificing developing-market sales, to defend their positions in developed markets. Nevertheless, the impact of MPS on export performance should differ across destinations.

**Table 11**

Supplier-side MPS and changes in suppliers - Transaction cost.

This table explores how the impact of supplier-side monetary policy shocks on supplier composition differs depending on the transaction cost. Panel A, B, and C use geographical distance, contiguosness, and legal origin as proxies for the distance between the firm and its suppliers, respectively. For all subsample analyses, the sample is divided by comparing the grouping variable to the yearly median. To save space, we only report the results on *Separation Rate* (odd columns) and *Termination Rate* (even columns). The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns.

| Panel A: Geographical distance |                        |                         |                        |                         |
|--------------------------------|------------------------|-------------------------|------------------------|-------------------------|
|                                | Close                  |                         | Distant                |                         |
|                                | <i>Separation Rate</i> | <i>Termination Rate</i> | <i>Separation Rate</i> | <i>Termination Rate</i> |
|                                | (1)                    | (2)                     | (3)                    | (4)                     |
| <i>SupplierMPS</i>             | 0.068<br>(0.125)       | 0.080<br>(0.124)        | 0.116*<br>(0.065)      | 0.113*<br>(0.061)       |
| Controls                       | Yes                    | Yes                     | Yes                    | Yes                     |
| Firm FE                        | Yes                    | Yes                     | Yes                    | Yes                     |
| Industry $\times$ Year FE      | Yes                    | Yes                     | Yes                    | Yes                     |
| City $\times$ Year FE          | Yes                    | Yes                     | Yes                    | Yes                     |
| Observations                   | 5,594                  | 5,594                   | 5,445                  | 5,445                   |
| $R^2$                          | 0.414                  | 0.412                   | 0.391                  | 0.393                   |
| Panel B: Contiguousness        |                        |                         |                        |                         |
|                                | Close                  |                         | Distant                |                         |
|                                | (5)                    | (6)                     | (7)                    | (8)                     |
|                                |                        |                         |                        |                         |
| <i>SupplierMPS</i>             | -0.030<br>(0.093)      | -0.022<br>(0.093)       | 0.149*<br>(0.083)      | 0.148*<br>(0.081)       |
| Controls                       | Yes                    | Yes                     | Yes                    | Yes                     |
| Firm FE                        | Yes                    | Yes                     | Yes                    | Yes                     |
| Industry $\times$ Year FE      | Yes                    | Yes                     | Yes                    | Yes                     |
| City $\times$ Year FE          | Yes                    | Yes                     | Yes                    | Yes                     |
| Observations                   | 5,692                  | 5,692                   | 5,380                  | 5,380                   |
| $R^2$                          | 0.425                  | 0.424                   | 0.380                  | 0.381                   |
| Panel C: Legal Origin          |                        |                         |                        |                         |
|                                | Close                  |                         | Distant                |                         |
|                                | (9)                    | (10)                    | (11)                   | (12)                    |
|                                |                        |                         |                        |                         |
| <i>SupplierMPS</i>             | 0.095<br>(0.134)       | 0.105<br>(0.133)        | 0.119*<br>(0.063)      | 0.116**<br>(0.059)      |
| Controls                       | Yes                    | Yes                     | Yes                    | Yes                     |
| Firm FE                        | Yes                    | Yes                     | Yes                    | Yes                     |
| Industry $\times$ Year FE      | Yes                    | Yes                     | Yes                    | Yes                     |
| City $\times$ Year FE          | Yes                    | Yes                     | Yes                    | Yes                     |
| Observations                   | 5,487                  | 5,487                   | 5,513                  | 5,513                   |
| $R^2$                          | 0.420                  | 0.419                   | 0.387                  | 0.387                   |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

We classify OECD members as developed countries and others as developing countries. The subsample results are reported in Panel B of Table 12. The significant decline in export value, quantity, and quality induced by MPS is observed only in exports to developing countries (columns (10)–(12)), while exports to developed countries remain largely unaffected (columns (7)–(9)). This contrast implies that when exposed to exogenous shocks in overseas sourcing, Chinese firms tend to protect established positions in developed countries by cutting export volume and even downgrading product quality to developing countries.

For an overall assessment of the impacts on firm performance, column (6) of Panel A shows that an increase in supplier-side MPS contributes to a significant slowdown in net profit growth. The above findings confirm the adverse real economic outcomes induced by supplier-side monetary policy shocks.

The analysis underscores that supplier switching is not frictionless. It reveals that heightened monetary policy shocks in supplier countries not only destabilize supply chains but also lead to tangible negative impacts on firms' real performance. This highlights the intricate web of international financial interdependence, where monetary policy changes in one country reverberate through global trading networks, ultimately affecting firms in countries with strict capital controls. These findings illuminate the substantial and far-reaching consequences of monetary policy shocks on the real economy, emphasizing the need for nuanced policy considerations in an interconnected global marketplace.

**Table 12**

Supplier-side MPS and real economic outcomes.

This table explores the real economic outcomes induced by supplier-side monetary policy shocks. Panel A uses the whole sample. The dependent variables are the value of import as a ratio of total operating cost (column (1)), import diversity (column (2)), export value (in log, column (3)), export quantity (column (4)), and export quality (column (5)), and net profit growth rate (column (6)). Import diversity is measured as one minus the import HHI, calculated at firm-HS code-import source country level. Export quality is inferred based on export quantity and price. The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Panel B differentiates between exports to developed (OECD members) and developing countries (the remaining countries). Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). Columns (1)–(2) and (6) are based on firm-year data and we control for firm, industry  $\times$  year, and city  $\times$  year fixed effects in all columns. Columns (3)–(5) and (7)–(12) use firm-HS code  $\times$  export destination country  $\times$  year data instead, and we control for firm, industry  $\times$  year, city  $\times$  year, 6-digit HS code  $\times$  year, and export destination country  $\times$  year fixed effects..

| Panel A.                  |                      |                        |                       |                        |                        |                          |
|---------------------------|----------------------|------------------------|-----------------------|------------------------|------------------------|--------------------------|
|                           | <i>Import share</i>  | <i>Import HHI</i>      | <i>Export Value</i>   | <i>Export quantity</i> | <i>Export quality</i>  | <i>Net profit growth</i> |
|                           | (1)                  | (2)                    | (3)                   | (4)                    | (5)                    | (6)                      |
| <i>SupplierMPS</i>        | −0.080***<br>(0.017) | −0.048**<br>(0.023)    | −0.362***<br>(0.097)  | −0.362***<br>(0.110)   | −0.196**<br>(0.097)    | −0.549*<br>(0.307)       |
| Controls                  | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| Firm FE                   | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| Industry $\times$ Year FE | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| City $\times$ Year FE     | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| HS6 $\times$ Year FE      | No                   | No                     | Yes                   | Yes                    | Yes                    | No                       |
| Country $\times$ Year FE  | No                   | No                     | Yes                   | Yes                    | Yes                    | No                       |
| Observations              | 11,910               | 11,910                 | 300,187               | 299,246                | 299,067                | 10,729                   |
| $R^2$                     | 0.723                | 0.797                  | 0.502                 | 0.656                  | 0.253                  | 0.650                    |
| Panel B.                  |                      |                        |                       |                        |                        |                          |
|                           | Developed Countries  |                        |                       | Developing Countries   |                        |                          |
|                           | <i>Export Value</i>  | <i>Export Quantity</i> | <i>Export Quality</i> | <i>Export Value</i>    | <i>Export Quantity</i> | <i>Export Quality</i>    |
|                           | (7)                  | (8)                    | (9)                   | (10)                   | (11)                   | (12)                     |
| <i>SupplierMPS</i>        | −0.152<br>(0.107)    | −0.130<br>(0.143)      | −0.093<br>(0.131)     | −0.510***<br>(0.157)   | −0.445***<br>(0.152)   | −0.383***<br>(0.142)     |
| Controls                  | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| Firm FE                   | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| Industry $\times$ Year FE | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| City $\times$ Year FE     | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| HS6 $\times$ Year FE      | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| Country $\times$ Year FE  | Yes                  | Yes                    | Yes                   | Yes                    | Yes                    | Yes                      |
| Observations              | 95,325               | 95,002                 | 94,973                | 202,213                | 201,603                | 201,477                  |
| $R^2$                     | 0.488                | 0.596                  | 0.350                 | 0.543                  | 0.697                  | 0.267                    |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

## 7. Conclusion

In this paper, we provide evidence on whether and how Chinese firms adapt supply chains in response to global monetary policy shocks, using detailed firm-to-firm supply chain data from the Factset Revere and international trade data from the Panjiva from 2011 through 2018. We start the analysis by constructing a firm-level exposure to monetary policy shocks based on its global supplier network and the variations of monetary policy across countries. We document five main findings.

First, firms are more likely to terminate suppliers, both in foreign and domestic markets, and establish fewer new foreign relationships when exposed to fierce monetary policy shocks (MPS). Second, firms diversify the risks of MPS by switching their suppliers from a group of high-interest-rate countries to the other of low-interest-rate countries. Particularly, when the interest rate of a foreign country increases, firms will shift suppliers from that country to other foreign markets but not choose to re-shore. Third, interest expenses and profitability of Chinese firms' suppliers are negatively affected by MPS from their countries, and the bad consequence is transmitted to customers through reduced trade credit and increased output prices. These provide explanations of why Chinese firms adapt supply chains. Also, disruption as a response to MPS is more likely to occur when the customer has greater relative strength in a supply chain, whereas the relative strength of the supplier does not produce differential impacts. Fourth, credit constraints, relationship strength, and trading costs impact firms' responses to MPS. Last, the adjustments of supply chains generate substantial costs and deteriorate firm profitability and export performance.

Our results overall show that global monetary shocks affect supply chain disruption and formation of Chinese firms through international trade networks, corroborating the real effects of monetary policy and the importance of production networks to transmit idiosyncratic and aggregate shocks in a cross-border setting. This paper has important implications for the restructuring of global supply chains in light of diversified monetary policies after the COVID-19 pandemic and increased geopolitical risk due to the China-US conflicts and the Russia-Ukraine War, among others.

## Appendix

Table A.1

Long run supplier MPS, supplier MPS Change, and supply chain adaptation.

This table uses alternative supplier-side MPS measures. The key independent variable in Columns (1)–(2) is *Longrun\_SupplierMPS*, where we use ten-year government bond yield instead of money market interest rate to construct MPS exposure. To capture change in monetary policy instead of the level, the key independent variable in columns (3)–(4) is the change in supplier-side MPS. The dependent variables are the changes in suppliers, including *Separation Rate* (columns (1) and (3)) and *Termination Rate* (columns (2) and (4)). Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We include firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                            | <i>Separation rate</i> | <i>Termination rate</i> | <i>Separation rate</i> | <i>Termination rate</i> |
|----------------------------|------------------------|-------------------------|------------------------|-------------------------|
|                            | (1)                    | (2)                     | (3)                    | (4)                     |
| <i>Longrun_SupplierMPS</i> | 0.087*<br>(0.051)      | 0.094*<br>(0.050)       |                        |                         |
| <i>SupplierMPS_Change</i>  |                        |                         | 0.080*<br>(0.046)      | 0.077*<br>(0.044)       |
| Controls                   | Yes                    | Yes                     | Yes                    | Yes                     |
| Firm FE                    | Yes                    | Yes                     | Yes                    | Yes                     |
| Industry $\times$ Year FE  | Yes                    | Yes                     | Yes                    | Yes                     |
| City $\times$ Year FE      | Yes                    | Yes                     | Yes                    | Yes                     |
| Observations               | 11,808                 | 11,808                  | 9,890                  | 9,890                   |
| $R^2$                      | 0.334                  | 0.332                   | 0.349                  | 0.348                   |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

Table A.2

Controlling for the impact of domestic economic environment.

This table provides additional evidence on controlling for the impact of domestic economic environment. In columns (1), (3), (4), (6), we use an updated supplier-side MPS focusing only on the foreign suppliers. Specifically, *SupplierMPS\_Foreign* is a weighted average foreign money market interest rate, where the weight is the share of import from each supplying country over total import. Furthermore, we control for the interaction terms of the share of imported inputs and year dummies in columns (2)–(3) and (5)–(6). The dependent variables are the changes in suppliers, including *Separation Rate* (columns (1)–(3)) and *Termination Rate* (columns (4)–(6)). Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We include firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                              | <i>Separation rate</i> |                    |                   | <i>Termination rate</i> |                    |                   |
|------------------------------|------------------------|--------------------|-------------------|-------------------------|--------------------|-------------------|
|                              | (1)                    | (2)                | (3)               | (4)                     | (5)                | (6)               |
| <i>SupplierMPS_Foreign</i>   | 0.013*<br>(0.007)      |                    | 0.014*<br>(0.007) | 0.013*<br>(0.007)       |                    | 0.014*<br>(0.007) |
| <i>SupplierMPS</i>           |                        | 0.142**<br>(0.070) |                   |                         | 0.141**<br>(0.068) |                   |
| ImportShare $\times$ Year FE | No                     | Yes                | Yes               | No                      | Yes                | Yes               |
| Controls                     | Yes                    | Yes                | Yes               | Yes                     | Yes                | Yes               |
| Firm FE                      | Yes                    | Yes                | Yes               | Yes                     | Yes                | Yes               |
| Industry $\times$ Year FE    | Yes                    | Yes                | Yes               | Yes                     | Yes                | Yes               |
| City $\times$ Year FE        | Yes                    | Yes                | Yes               | Yes                     | Yes                | Yes               |
| Observations                 | 2,925                  | 11,910             | 2,925             | 2,925                   | 11,910             | 2,925             |
| $R^2$                        | 0.447                  | 0.333              | 0.449             | 0.450                   | 0.331              | 0.452             |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.



Table A.3

Controlling for the import share - Firm-country-year sample.

This table provides additional evidence on the firm-country-year panel (as in Table 5) by including the import share from the corresponding country as an additional control variable. The regression model is as follows:  $Supplier\_Change_{ikt} = \beta_1 Rate_{kt} + Controls + Firm_i + Country_k + Industry_i \times Year_t + City_i \times Year_t + \epsilon_{ikt}$ .  $Rate_{kt}$  is the money market interest rate of country  $k$  in year  $t$ . The dependent variables are the changes in suppliers, including *Separation Rate* (columns (1)–(2)), *Entry Rate* (columns (3)–(4)). Control variables include total assets in log ( $Ln(Asset)$ ), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age ( $Ln(Age)$ ), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for the bilateral real exchange rate of country  $k$  in year  $t$ . We include firm and country fixed effects in all columns. Odd columns include year fixed effect in odd columns, while even columns control for industry  $\times$  year, city  $\times$  year fixed effects..

|                           | Separation rate    |                   | Entry rate       |                   |
|---------------------------|--------------------|-------------------|------------------|-------------------|
|                           | (1)                | (2)               | (3)              | (4)               |
| <i>Rate</i>               | 0.004**<br>(0.001) | 0.003*<br>(0.002) | 0.003<br>(0.005) | −0.001<br>(0.006) |
| <i>ImportShare</i>        | 0.002<br>(0.013)   | −0.002<br>(0.021) | 0.021<br>(0.035) | 0.073<br>(0.054)  |
| Controls                  | Yes                | Yes               | Yes              | Yes               |
| Firm FE                   | Yes                | Yes               | Yes              | Yes               |
| Country FE                | Yes                | Yes               | Yes              | Yes               |
| Year FE                   | Yes                | No                | Yes              | No                |
| Industry $\times$ Year FE | No                 | Yes               | No               | Yes               |
| City $\times$ Year FE     | No                 | Yes               | No               | Yes               |
| Observations              | 13,202             | 12,963            | 13,202           | 12,963            |
| <i>R</i> <sup>2</sup>     | 0.132              | 0.283             | 0.097            | 0.226             |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

Table A.4

Robustness check - Alternative measures.

This table provides robustness checks by using alternative measures for the changes in suppliers. The dependent variables in columns (1)–(2) are the number of suppliers (in log) and the changes in the number of suppliers, respectively. Columns (3)–(4) and (5)–(6) explore the impacts on supplier exits and entries, using the log numbers of supplier exits and entries. The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log ( $Ln(Asset)$ ), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age ( $Ln(Age)$ ), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                           | $Ln(SuppNo.)$      | $\Delta SuppNo.$    | $Ln(Separated\ SuppNo.)$ | $Ln(Terminated\ SuppNo.)$ | $Ln(Entered\ SuppNo.)$ | $Ln(Debut\ SuppNo.)$ |
|---------------------------|--------------------|---------------------|--------------------------|---------------------------|------------------------|----------------------|
|                           | (1)                | (2)                 | (3)                      | (4)                       | (5)                    | (6)                  |
| <i>SupplierMPS</i>        | −0.127*<br>(0.074) | −0.719**<br>(0.325) | 0.073*<br>(0.040)        | 0.076**<br>(0.038)        | −0.093<br>(0.065)      | −0.090<br>(0.065)    |
| Controls                  | Yes                | Yes                 | Yes                      | Yes                       | Yes                    | Yes                  |
| Firm FE                   | Yes                | Yes                 | Yes                      | Yes                       | Yes                    | Yes                  |
| Industry $\times$ Year FE | Yes                | Yes                 | Yes                      | Yes                       | Yes                    | Yes                  |
| City $\times$ Year FE     | Yes                | Yes                 | Yes                      | Yes                       | Yes                    | Yes                  |
| Observations              | 11,910             | 11,910              | 11,910                   | 11,910                    | 11,910                 | 11,910               |
| <i>R</i> <sup>2</sup>     | 0.676              | 0.331               | 0.477                    | 0.470                     | 0.506                  | 0.503                |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

**Table A.5**

Robustness check - The customs data.

This table provides robustness checks by using alternative data source - the Customs data. We run similar regression as in columns (2) and (4) of Table 2 using the Customs data. The dependent variables capture changes in suppliers, including *Separation Rate* and *Termination Rate*. The key independent variable is the *SupplierMPS*. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                           | <i>Separation rate</i> | <i>Termination rate</i> |
|---------------------------|------------------------|-------------------------|
|                           | (1)                    | (2)                     |
| <i>SupplierMPS</i>        | 0.029*<br>(0.017)      | 0.029*<br>(0.016)       |
| Controls                  | Yes                    | Yes                     |
| Firm FE                   | Yes                    | Yes                     |
| Industry $\times$ Year FE | Yes                    | Yes                     |
| City $\times$ Year FE     | Yes                    | Yes                     |
| Observations              | 17,734                 | 17,734                  |
| $R^2$                     | 0.329                  | 0.324                   |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

**Table A.6**

Robustness check - Excluding certain firms.

This table provides robustness checks by excluding firms with certain characteristics from the sample. Columns (1)–(2) present the results excluding firms established in less than 10 years and columns (3)–(4) exclude large firms with size among the top decile. To save space, we only report the results on *Separation Rate* (Columns (1) and (3)) and *Termination Rate* (Columns (2) and (4)). The main variable of interest is *SupplierMPS*, the average money market interest rate of the suppliers, weighted by the input share. Control variables include real effective exchange rate (*REER*), total assets in log (*Ln(Asset)*), debt to asset ratio (*Leverage*), return on assets (*ROA*), receivable ratio (*Receivable*), revenue growth rate (*RevGrowth*), net cash flow per share (*CashFlow*), firm age (*Ln(Age)*), and total holding ratio of top 5 shareholders (*Top5Ratio*). We control for firm, industry  $\times$  year, city  $\times$  year fixed effects in all columns..

|                           | Excluding young firms (Age < 10) |                         | Excluding large firms (Size > p90) |                         |
|---------------------------|----------------------------------|-------------------------|------------------------------------|-------------------------|
|                           | <i>Separation rate</i>           | <i>Termination rate</i> | <i>Separation rate</i>             | <i>Termination rate</i> |
|                           | (1)                              | (2)                     | (3)                                | (4)                     |
| <i>SupplierMPS</i>        | 0.106*<br>(0.058)                | 0.112**<br>(0.056)      | 0.125*<br>(0.065)                  | 0.125*<br>(0.064)       |
| Controls                  | Yes                              | Yes                     | Yes                                | Yes                     |
| Firm FE                   | Yes                              | Yes                     | Yes                                | Yes                     |
| Industry $\times$ Year FE | Yes                              | Yes                     | Yes                                | Yes                     |
| City $\times$ Year FE     | Yes                              | Yes                     | Yes                                | Yes                     |
| Observations              | 10,860                           | 10,860                  | 10,591                             | 10,591                  |
| $R^2$                     | 0.348                            | 0.347                   | 0.343                              | 0.342                   |

\*, \*\*, \*\*\* stands for significance at 10%, 5%, 1% level, respectively. Standard errors are reported in the parentheses below the estimated parameters. Standard errors clustered at firm level.

## Data availability

Data will be made available on request.

## References

- A, Petersen Mitchell, & Raghuram, G. Rajan (1997). Trade credit: Theories and evidence. *The Review of Financial Studies*, 10(3), 661–691.
- Aeimit, Lakdawala, Moreland, Timothy, & Schaffer, Matthew (2021). The international spillover effects of US monetary policy uncertainty. *Journal of International Economics*, 133, Article 103525.
- Alfaro, Laura, Brussevich, Mariya, Minoiu, Camelia, & Presbitero, Andrea F. (2025). *Bank financing of global supply chains. national bureau of economic research: Technical Report*.
- Alfaro, Laura, García-Santana, Manuel, & Moral-Benito, Enrique (2021). On the direct and indirect real effects of credit supply shocks. *Journal of Financial Economics*, 139(3), 895–921.

- Amiti, Mary, Itskhoki, Oleg, & Konings, Jozef (2014). Importers, exporters, and exchange rate disconnect. *American Economic Review*, 104(7), 1942–1978.
- Anna, M. Costello (2020). Credit market disruptions and liquidity spillover effects in the supply chain. *Journal of Political Economy*, 128(9), 3434–3468.
- B, Hendricks Kevin, & Vinod, R. Singhal (2005). An empirical analysis of the effect of supply chain disruptions on long-run stock price performance and equity risk of the firm. *Production and Operations Management*, 14(1), 35–52.
- Baier, Scott L., Kerr, Amanda, & Yotov, Yoto V. (2018). Gravity, distance, and international trade. In *handbook of international trade and transportation* (pp. 15–78). Edward elgar publishing.
- Balboni, Clare, Boehm, Johannes, & Waseem, Mazhar (2024). Firm adaptation in production networks: Evidence from extreme weather events in Pakistan. Working Paper, Structural Transformation and Economic Growth (STEG)\*..
- Ben, Bernanke, & Blinder, Alan S. (1992). The federal funds rate and the transmission of monetary policy. *American Economic Review*, 82(4), 901–921.
- Ben, Bernanke, & Gertler, Mark (1995). Inside the black box: the credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4), 27–48.
- Bigio, Saki, & Jennifer, La' O. (2016). Distortions in production networks. National bureau of economic research. *Technical Report*.
- Blomhoff, Holm Martin, Paul, Pascal, & Tischbirek, Andreas (2021). The transmission of monetary policy under the microscope. *Journal of Political Economy*, 129(10), 2861–2904.
- Boehm, Christoph E., Flaaen, Aaron, & Pandalai-Nayar, Nitya (2019). Input linkages and the transmission of shocks: Firm-level evidence from the 2011 Tōhoku earthquake. *Review of Economics and Statistics*, 101(1), 60–75.
- Cetorelli, Nicola, & Gambera, Michele (2001). Banking market structure, financial dependence and growth: International evidence from industry data. *The Journal of Finance*, 56(2), 617–648.
- Charoenwong, Ben, Han, Miaoze, & Wu, Jing (2023). Trade and foreign economic policy uncertainty in supply chain networks: Who comes home? *Manufacturing & Service Operations Management*, 25(1), 126–147.
- Chen, Sihao (2023). International production networks and the propagation of financial shocks. Available at SSRN 3815081.
- Choi, Jaerim, Hyun, Jay, & Park, Zihao (2024). Bound by ancestors: Immigration, credit frictions, and global supply chain formation. *Journal of International Economics*, 147, Article 103855.
- Chuoyan, Dong Maggie, Huang, Qianqian, & Liu, Zhiqiang (2022). Adjusting supply chain involvement in countries with politician turnover: A contingency framework. *Journal of Operations Management*, 68(8), 824–854.
- Cloyne, James, Ferreira, Clodomiro, Froemel, Maren, & Surico, Paolo (2023). Monetary policy, corporate finance, and investment. *Journal of the European Economic Association*, 21(6), 2586–2634.
- Daniela, Fabbri, & Klapper, Leora F. (2016). Bargaining power and trade credit. *Journal of Corporate Finance*, 41, 66–80.
- Dass, Nishant, Kale, Jayant R., & Nanda, Vikram (2015). Rade credit, relationship-specific investment, and product market power. *Review of Finance*, 19(5), 1867–1923.
- Dunlop John, T., & Benjamin, Higgins (1942). Bargaining power and market structures. *Journal of Political Economy*, 50(1), 1–26.
- Durante, Elena, Ferrando, Annalisa, & Vermeulen, Philip (2022). Monetary policy, investment and firm heterogeneity. *European Economic Review*, 148, Article 104251.
- E, Givens Gregory, & Robert, R. Reed (2018). Monetary policy and investment dynamics: Evidence from disaggregate data. *Journal of Money, Credit and Banking*, 50(8), 1851–1878.
- Elias, Albagli, Ceballos, Luis, Claro, Sebastian, & Romero, Damian (2019). Channels of US monetary policy spillovers to international bond markets. *Journal of Financial Economics*, 134(2), 447–473.
- Emi, Nakamura, & Steinsson, Jon (2018). High-frequency identification of monetary non-neutrality: The information effect. *The Quarterly Journal of Economics*, 133(3), 1283–1330.
- Gene, M. Grossman, Helpman, Elhanan, & Redding, Stephen J. (2024). When tariffs disrupt global supply chains. *American Economic Review*, 114(4), 988–1029.
- Gerard, Weisbuch, Kirman, Alan, & Herreiner, Dorothea (2000). Market organisation and trading relationships. *The Economic Journal*, 110(463), 411–436.
- Grossman, Gene M., Helpman, Elhanan, & Lhuillier, Hugo (2023). Supply chain resilience: Should policy promote international diversification or reshoring? *Journal of Political Economy*, 131(12), 3462–3496.
- Grossman Sanford, J., & Oliver, D. Hart (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. *Journal of Political Economy*, 94(4), 691–719.
- Guillermo, Marshall (2020). Search and wholesale price discrimination. *Rand Journal of Economics*, 51(2), 346–374.
- Gurkaynak, Refet S., Sack, Brian, & Swanson, Eric T. (2005). Do actions speak louder than words? The response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*.
- Hiroiyuki, Kasahara, & Rodrigue, Joel (2008). Does the use of imported intermediates increase productivity? Plant-level evidence. *Journal of Development Economics*, 87(1), 106–118.
- Ioannidou, Vasso, Ongena, Steven, & Peydró, José-Luis (2015). Monetary policy, risk-taking, and pricing: Evidence from a quasi-natural experiment. *Review of Finance*, 19(1), 95–144.
- James, E. Anderson, & Eric, Van Wincoop (2003). Gravity with gravitas: A solution to the border puzzle. *American Economic Review*, 93(1), 170–192.
- Jian, Wang, & Wang, Xiao (2015). Benefits of foreign ownership: Evidence from foreign direct investment in China. *Journal of International Economics*, 97(2), 325–338.
- Julian, Di Giovanni, & Hale, Galina (2022). Stock market spillovers via the global production network: Transmission of US monetary policy. *The Journal of Finance*, 77(6), 3373–3421.
- Kalina, Manova, Wei, Shang-Jin, & Zhang, Zhiwei (2015). Firm exports and multinational activity under credit constraints. *Review of Economics and Statistics*, 97(3), 574–588.
- Kalina, Manova, & Yu, Zhihong (2017). Multi-product firms and product quality. *Journal of International Economics*, 109, 116–137.
- Kalina, Manova, & Zhang, Zhiwei (2012). Export prices across firms and destinations. *The Quarterly Journal of Economics*, 127(1), 379–436.
- Kashyap Anil, K., & Jeremy, C. Stein (2000). What do a million observations on banks say about the transmission of monetary policy? *American Economic Review*, 90(3), 407–428.
- Kenneth, N. Kuttner (2001). Monetary policy surprises and interest rates: Evidence from the fed funds futures market. *Journal of Monetary Economics*, 47(3), 523–544.
- Khanna, Gaurav, Morales, Nicolas, & Pandalai-Nayar, Nitya (2022). *Supply chain resilience: evidence from Indian firms*. National bureau of economic research: Technical Report.
- Klaus, E. Meyer, Li, Chengguang, & Schotter, Andreas P. J. (2020). Managing the MNE subsidiary: Advancing a multi-level and dynamic research agenda. *Journal of International Business Studies*, 51, 538–576.
- Krauthaim, Sebastian (2012). Heterogeneous firms, exporter networks and the effect of distance on international trade. *Journal of International Economics*, 87(1), 27–35.
- Laurent, Frésard, Hege, Ulrich, & Phillips, Gordon (2017). Extending industry specialization through cross-border acquisitions. *The Review of Financial Studies*, 30(5), 1539–1582.
- Li, Yao Amber, Lu, Lingfei, Wei, Shang-Jin, & Yao, Jingbo (2024). Global monetary policy shocks, financial frictions, and export prices. Working Paper.
- Lin, Shu, & Ye, Haichun (2018). Foreign direct investment, trade credit, and transmission of global liquidity shocks: Evidence from Chinese manufacturing firms. *The Review of Financial Studies*, 31(1), 206–238.

- Manova, Kalina (2013). Credit constraints, heterogeneous firms, and international trade. *Review of Economic Studies*, 80(2), 711–744.
- Manuel, Adelino, Ferreira, A., Giannetti, Mariassunta, & Pires, Pedro (2023). Trade credit and the transmission of unconventional monetary policy. *The Review of Financial Studies*, 36(2), 775–813.
- Matias, Berthelon, & Freund, Caroline (2008). On the conservation of distance in international trade. *Journal of International Economics*, 75(2), 310–320.
- Matteo, Ciccarelli, Maddaloni, Angela, & Peydró, José-Luis (2015). Trusting the bankers: A new look at the credit channel of monetary policy. *Review of Economic Dynamics*, 18(4), 979–1002.
- Matthew, Elliott, Golub, Benjamin, & Leduc, Matthew V. (2022). Supply network formation and fragility. *American Economic Review*, 112(8), 2701–2747.
- M.C., Pankratz Nora, & Christoph, M. Schiller (2024). Climate change and adaptation in global supply-chain networks. *The Review of Financial Studies*, 37(6), 1729–1777.
- Michael, Gofman, & Wu, Youchang (2022). Trade credit and profitability in production networks. *Journal of Financial Economics*, 143(1), 593–618.
- Minetti, Raoul, Murro, Pierluigi, Rotondi, Zeno, & Zhu, Susan Chun (2019). Financial constraints, firms' supply chains, and internationalization. *Journal of the European Economic Association*, 17(2), 327–375.
- Mishkin, Frederic S. (2009). Is monetary policy effective during financial crises? *American Economic Review*, 99(2), 573–577.
- Obstfeld, Maurice, & Rogoff, Kenneth (2000). The six major puzzles in international macroeconomics: Is there a common cause? *NBER Macroeconomics Annual*, 15, 339–390.
- Pablo, Ottonello, & Winberry, Thomas (2020). Financial heterogeneity and the investment channel of monetary policy. *Econometrica*, 88(6), 2473–2502.
- Pamina, Koenig, & Poncet, Sandra (2022). The effects of the rana plaza collapse on the sourcing choices of French importers. *Journal of International Economics*, 137, Article 103576.
- Paulo, Bastos, Silva, Joana, & Verhoogen, Eric (2018). Export destinations and input prices. *American Economic Review*, 108(2), 353–392.
- Porta, Rafael La, Lopez-de-Silanes, Florencio, & Shleifer, Andrei (2008). The economic consequences of legal origins. *Journal of Economic Literature*, 46(2), 285–332.
- Raghuram, Rajan, & Zingales, Luigi (1998). Financial dependence and growth. *American Economic Review*, 88(3), 559–586.
- Senay, Agca, Babich, Volodymyr, Birge, R., & Wu, Jing (2022). Credit shock propagation along supply chains: Evidence from the CDS market. *Management Science*, 68(9), 6506–6538.
- Silvia, Miranda-Agrippino, & Rey, Helene (2020). US monetary policy and the global financial cycle. *Review of Economic Studies*, 87(6), 2754–2776.
- Tan, Weiqiang, Wang, Wenming, & Zhang, Wenlan (2024). The effects of terrorist attacks on supplier–customer relationships. *Production and Operations Management*, 10591478231224920.
- Tatjana, Dahlhaus, & Vasishtha, Garima (2020). Monetary policy news in the US: Effects on emerging market capital flows. *Journal of International Money and Finance*, 109, Article 102251.
- Thomas, Chaney (2018). The gravity equation in international trade: An explanation. *Journal of Political Economy*, 126(1), 150–177.
- Tony, Zhang (2022). Monetary policy spillovers through invoicing currencies. *The Journal of Finance*, 77(1), 129–161.
- Vasco, M. Carvalho, Nirei, Makoto, Saito, U., & Tahbaz-Salehi, Alireza (2021). Supply chain disruptions: Evidence from the great east Japan earthquake. *The Quarterly Journal of Economics*, 136(2), 1255–1321.
- Vicente, Cunat (2007). Trade credit: Suppliers as debt collectors and insurance providers. *The Review of Financial Studies*, 20(2), 491–527.
- Viral, Acharya, & Xu, Zhaoxia (2017). Financial dependence and innovation: The case of public versus private firms. *Journal of Financial Economics*, 124(2), 223–243.